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Quantitative Analysis for Business (PRC - 02)

Last Assignment

For

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by

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1 Linear and Quadratic equations

1) Total Revenue earned by Mr. X is Rs. 550,000 and variable cost & Fixed Cost incurred by Mr. X is Rs. 270,000 & Rs. 100,000 respectively. Calculate the total contribution earned by Mr. X is?

a) Rs. 270000	b) Rs 370000	**c) Rs. 280,000**	d) Rs. 180,000

2) Which of the following is irrational?

a) √(4/9)	b) 4/5	c) √7	d) √(81)

3) The total cost of producing 12 units is Rs. 1,500 and the total cost of producing 18 units is Rs. 2,700. Find the variable cost per unit?

a) 125	**b) 200**
c) 150	d) 140

4) Find the value of x:

$$\frac{(x+1)}{(x-1)} = \frac{4}{5} \quad and \quad \frac{(x-1)}{(x+1)} = \frac{1}{2}$$

a) (9,5)	b) (-7,-9)
c) (7,9)	**d) (-9,3)**

5) 9 years ago the age of father was 3 times of his son's age. After 9 years it will be twice the age of his son. The present ages of father and son are:

a) 48 and 20	b) 51 and 21
c) 63 and 27	d) 57 and 23

6) 6 years ago the age of father was 3 times of his son's age. After 9 years it will be twice the age of his son. The present ages of father and son are:

a) 48 and 20	**b) 51 and 21**
c) 54 and 22	d) 57 and 23

7) Seven years ago the age of a father was thrice of his son. After 7 yean the age of the father will be twice that of his son. The present ages of the father and the son are?

a) (69,23)	b) (85,39)	c) (78,32)	**d) None of these**

8) A line passes through a point (2,3) and cuts at x-intercept of 3. Find equation of that line.

a) Y= 3x+9	b) Y= 3x- 9
c) Y= - 9x+3	**d) Y= - 3x+9**

9) 5 years ago the age of father was 3 times of his son's age. After 7 years it will be twice the age of his son. The present age of father and son is:

a) 53 and 21	**b) 41 and 17**
c) 50 and 20	d) 47 and 19

10) The equation representing a straight line is 7y=11x+3. The y-intercept is:

a) 3/7	b) 11/3
c) 7/3	d) 11/7

11) The equation representing a straight line is 4y=5x+8. The slope of line is:

a) 1.25	b) 5
c) 0.8	d) 0.5

12) Which of the following equations is not linear?

a) $y = 2x^2$	b) X+9=0
c) $x - \frac{y}{5} + 20 = 0$	d) Y=2x-5

13) Kamran and Salman invested in a business. The sum of investment of Kamran and seven times the investment of Salman amounts to Rs 18 million. Difference between thrice the investment of Kamran and twice the investment of Salman is Rs 8 million. Amount invested by Kamran and Salman is:

a) Rs 11 and 1 million respectively	b) Rs 7.5 and 1.5 million respectively
c) Rs 4 and 2 million respectively	d) Rs 8 million each.

14) Saad and Ali invested in a business. The sum of investment of Saad and seven times the investment of Ali amounts to Rs 18 million. Difference between thrice the investment of Saad and twice the investment of Ali is Rs 8 million. Amount invested by Saad and Ali is:

a) Rs 11 and 1 million respectively	b) Rs 7.5 and 1.5 million respectively
c) Rs 4 and 2 million respectively	d) Rs 8 million each.

15) Sum of thrice of Arif and twice of Ali is 18,000 and difference of thrice of Arif and twice of Ali is 15,000. Howmuch amount Arif and Ali have?

a) 5500, 750	b) 3000,750
c) 4500,750	d) 2500,750

16) Hamid and Sajid invested in a business. The sum of investment of Hamid and seven times the investment of Sajid amounts to Rs 18 million. Difference between thrice the investment of Hamid and twice the investment of Sajid is Rs 8 million. Amounts invested by Hamid and Sajid is:

a) Rs 11 and Rs 1 million respectively
b) Rs 7.5 million and Rs 1.5 million respectively
c) Rs 4 and Rs 2 million respectively.
d) Rs 8 million each.

17) Kamran and Salman invested in a business. The sum of investment of Kamran and seven times the investment of Salman amounts to Rs 18 million. Difference between thrice the investment of Kamran and twice the investment of Salman is Rs 8 million. Amounts invested by Kamran and Salman is:

a) Rs 11 and Rs 1 million respectively
b) Rs 7.5 million and Rs 1.5 million respectively
c) Rs 4 and Rs 2 million respectively.
d) Rs 8 million each.

18) A two-digit number is equal to "7 times the sum of its digits" and "21 times equal to the difference of its digits". If xy represents the required number, solving which of the following simultaneous equations could lead to that number?

a) 7y– xy =7x and 21x +xy=21y	b) 7x+xy=7y and 21x – xy=21y
c) 7x– xy = 7y and $x+\frac{xy}{21}=y$	**d) 7x – xy= –7y and 21y +xy=21x**

19) A two-digit number is equal to "4 times the sum of its digits" and "12 times equal to the difference of its digits." If xy represents the required number, solving which of the following simultaneous equations could lead to that number?

a) 4x+xy=4y and 12x – xy=12y	b) Xy – 4y =4x and 12y – xy=12x
c) Xy – 4x = 4y and xy +12y =12x	**d)** Xy – 4x = 4y and xy – 12x =12y

20) A line passes through (3,2). It has an X-intercept value which is thrice the y-intercept value. The slope intercept form of equation of the straight line is:

a)	b)
c)	**d)**

21) Which of the following equation is not linear?

a)	Y=2x-5	b)	$x-\frac{y}{5}+20\neq 0$
c)	**Y=2x^2**	d)	None of these

22) Which of the following statements is correct in respect of the equation? X+2Y+3=0

a)	The value of the intercept on the y-axis is 3
b)	**The value of the intercept on x-axis is –3**
c)	The slope of the line is 1
d)	The degree of the equation is 0

23) If a negative slope line passes through a point (4,8) and x intercept is ¼ of the y-intercept. Find equation of straight line

a) 4x+y=24	b) 4x – y=24
c) X+4y=24	d) X – 4y=24

24) Which of the following statements is correct in respect of the equation? 5X+2Y-10=0

a)	The value of the intercept on the y-axis is 2
b)	The value of the intercept on x-axis is –5

c)	The slope of the line is 2.5
d)	**The degree of the equation is 1**

25) A company has Revenue function R=9x and cost function $C = -0.09x^2 + 2x + 5{,}000$. You are required to determine Break-even point (in units) of this company.

a) 500 units	b) 700 units
c) 200 units	d) 400 units.

2 Coordinate System and Its Application

26) Which **TWO** of the following are quadratic equations?

a) $\frac{3}{5}x-12=2x$	b) $2x-11=5x$
c) $\sqrt{3}x^2-12x=15$	d) $5x^2-7=2x$

27) You are given an equation $15x-\frac{650}{x}=15$, Find the value of x?

a) - 6.1 & 7 1	b) -7 1 & 6.1	c) 6.1 & 3.1	d) -6 1 & -3.1

28) If sales value of x unit is 16.8x and the cost of sales is $-0.09x^2+1.8x+5000$ then the number of units at break-even will be?

a) 1200	b) 800	c) 333	**d) 167**

29) $25x^2-\frac{500}{x}=25x$. Find the value of x?

a) -4 & 5 b) 4 & -5 c) 6 & 3 **d) none of these**

30) $\sqrt{x^4-18x^2+81}=0$ which of the following represent the roots of the equation?

a) $\pm 2\sqrt{3}$	**b) ±3**
c) ±9	d) $\pm\sqrt{3}$

31) Which TWO of the following are not quadratic equations?

a) $\frac{3}{5}x-12=2x$	b) $2x-11=5x$
c) $\sqrt{3}x^2-12x=15$	d) $5x^2-7=2x$

32) Which one is not a quadratic equation?

a) 3x+4 =7 b) $2x^2 = 9$ c) $9x^2 + 2x = -3$ d) $5x^2 + 8x = 9$

33) Which of the following is not a quadratic equation?

a) $5x^2-7=2x$	**b) $2x-11=15$**
c) $\sqrt{3}x^2-12x=15$	d) $\frac{3}{5}x-12=12x^2$

34) Which of the following values of x will satisfy the equation? $x+10 = 11x^2 - x+1$

a) 1 and 0.818	b) – 1 and –0.818
c) 1 and –0.818	d) –1 and 0.818

35) Which of the following statements is correct for the equation $3x^2+5x=9$?

a) Coefficient of x is 2	**b) Constant = -9**
c) The equation contains two variables	d) It is a linear equation

36) A U-shaped curve:

a) Departs from symmetry and the frequencies tend to pileup at one or the other end of the curve	b) Represent values that are at equal distance from a central maximum value
c) Has the maximum frequencies occurring at both ends of the range and a minimum frequency towards the centre	d) Has frequencies that run up to a maximum at one end of the range

37) A U-shaped curve:

a) **Has the maximum frequencies occurring at both ends of the range and a minimum frequency towards the center**	b) Has frequencies that run up to a maximum at one end of the range
c) Represent values that are at equal distance from a central maximum value	d) Departs from symmetry and the frequencies tend to pileup at one or the other end of the curve

3 Mathematical progression

38) If Ahmad saves Rs. 1,000 in the first month and 500 more every month, how much time he needs to save Rs. 45,000

a) 9	**b) 12**	c) 8	d) 6

39) Basheer saves Rs. 105,000 in one year. if his first instalment was Rs. 500, how much more Rs. X will be added monthly to accumulate the above amount?

a) 500	b) 1,000	**c) 1,500**	d) 2,000

40) If someone invest Rs 6,000 as 1st instalment and increase by Rs.300 per month in each of next instalment for 24 months which of the following is true

1. Rs. 119,900 is the total value received after 13th instalment
2. The value for 13th instalment is Rs. 9,900

a) Only statement (1) is correct	b) Only statement (2) is correct
c) Both statements are correct	d) **Both statements are not correct**

41) Mr. Ali has borrowed X amount of money. He agreed to pay the installment 1.2 times than the last one. If his fifth installment is Rs. 51,389, what was his first installment?

a) 51,384.2	b) **24,782.5**
c) 20,652.1	d) Cannot be calculated

42) If a person invests 2 million and withdraw 10% each month then how much time it will take to end of investment?

a) 5 months	b) 15 months
c) 10 months	d) 20 months

43) XYZ and company has developed a new product which would earn a revenue of Rs. 80 million during the first year. Therefore, the revenue would decline by 20% each year. The company would be able to earn a revenue of Rs 400 million over the life of the product.

44) Mr. Adeel saved Rs x in January, then each subsequent month he saved Rs 100 more than the previous month. If his total savings at the end of December stood at Rs 16,200 how much did he save in January?

a) 700	**b) 800**
c) 900	d) 1,000

45) Ali borrowed Rs. 1,500,000 on first year he returned Rs. 80,000 and then he increases his instalment by 1.5 times of the previous instalment every year. In how many years he will be able to return the loan?

a) 9 years	b) 8 year	c) 10 years	**d) 6 years**

$$S = \frac{a\,(r^n - 1)}{r - 1}$$

$$1500000 = \frac{80000(1.5^n - 1)}{1.5 - 1}$$

By solving from calculator, we get "n" = 5.77= 6 years

46) Ali borrowed Rs. 1,500,000 on first year he returned Rs. 80,000 and then he increases his instalment by 1.2 times of the previous instalment every year. In which years he will be able to return the loan?

a) 9th years	b) 8th year	c) 10th years	d) 6th years

$$S = \frac{a(r^n - 1)}{r - 1}$$

$$1{,}500{,}000 = \frac{80{,}000(1.2^n - 1)}{1.2 - 1}$$

By solving from calculator, we get r = 8.54 = 9th years

47) Which of the following pairs of values cannot form part of a Geometric Progression?

a) (6 and -6)	b) $\sqrt{5}$ *and* $1/\sqrt{7}$
c) 1,000,000 and 0	d) **All of the above**

48) XYZ and Company has developed a new product which would earn a revenue of Rs 90 million during the first year. Thereafter, the revenue would decline by 10% each year. Calculate the revenue that the company would be able to earn over the life of the product

a) 450 million	b) 500 million
c) 900 million	d) 1 billion

Answer

$$S = \frac{a}{1-r} = \frac{90}{1-0.9} = 900\ million$$

49) ABC and Company has developed a new product which would earn a revenue of Rs 80 million during the first year. Thereafter, the revenue would decline by 20% each year. Calculate the revenue that the company would be able to earn over the life of the product

a) 400 million	b) 500 million
c) 800 million	d) 100 million

Answer

$$S = \frac{a}{1-r} = \frac{80}{1-0.8} = 400\ million$$

50) Ali receives a total of Rs. 288,000 in two years in the form of monthly installments. If he receives 1000 more than the previous month calculate the first installment?

a) Rs 7,500	b) Rs 6,500	**c) Rs 500**	d) None of these

51) Ali has to pay Rs. 1500000 in installments. If he pays the first installment of Rs. 80000 and increase everyinstallment by 1.2, then in which year he will pay the amount?

a) 9^{th} years	b) 8^{th} year	c) 10^{th} years	d) 11^{th} years

52) If Matti save Rs. 426000 monthly in two years. If he saves 500 in first month by x. Find the amount of lastinstallment.

a) 40000 **b) 35000** c) 36000 d) 33000

4 Linear Programming

53) A company makes and sells two products X and Y. the related information is as follows:

	X	Y
Contribution per unit	150	320
Maximum sales	5000	7000
Direct Labour per unit	3	2
Machine per unit	4	1

A total of 25,000 direct labour hours and 20,000 machine hours are available per month.

The objective function (Z) and set of constraints represents the above situation would be:

a) $Z=150x+320y$ $x \leq 5,000$ and $y \leq 7,000$ $3x+2y \leq 25,000$ $4x+y \geq 20,000$ $X,y \geq 0$	**b) $Z=150x+320y$** **$x \leq 5,000$ and $y \leq 7,000$** **$3x+2y \leq 25,000$** **$4x+y \leq 20,000$** **$x,y \geq 0$**
c) $Z=150x+320y$ $x \geq 5,000$ and $y \leq 7,000$ $3x+2y \leq 25,000$ $4x+y \leq 20000$ $X,y \geq 0$	d) $Z=150x+320y$ $x \leq 5,000$ and $y \geq 7,000$ $3x+2y \leq 25,000$ $4x+y \leq 20000$ $X,y \geq 0$

54) The profit for product x & y is Rs. 120 per unit and Rs. 80 per unit respectively.
Find the maximum profit under the given constraints:
$4x+5y \leq 51,000$
$7x+4y \leq 56,000$

a. 960,000	**b. 1,040,000**	c. 1,120,000	d. 1,230,000

55) A diversified sports company make bats and footballs.

Product Name	**Sales**	**Cost**
Bat	1,500	1,100
football	3,000	2,300

Manufacturing times for bat and football are 2 and 4 labour hours respectively. The company has 45 workers which can work 200 hours in a month the funds available for manufacturing are Rs. 5 million. Raw material is available as required. Determine which of the following constraints are correct?

a) $Z=1,500B+3,000F$ $2B+4F \leq 9,000$ $1,100B+2,300F \geq 5,000,000$	b) $Z=1500B+3,000F$ $4B+2F \geq 9,000$ $1,100B+2,300F \leq 5,000,000$
c) $Z=1,500B+3,000F$ **$2B+4F \leq 9,000$** **$1,100B+2,300F \leq 5,000,000$**	d) $Z=1500B+3000F$ $2B+4F \leq 9000$ $2300B+1100F \leq 5000000$

56) A company earns profit of Rs 250 and Rs 375 per unit on product X and Y respectively. Find the maximum profit that the company could earn if the company is subject to the following constraints:

$$4x+2y \leq 25,000$$
$$3x+2y \leq 20,000$$

a) Rs 1,666,667	b) Rs 2,187,500
c) Rs 3,750,000	d) Rs 4,687,500

57) A factory produces two products x and y. Each product passes through two departments A and B which have a capacity of 1120 hours and 1400 hours The Product X requires 4 hours in department A and 7 hours in department B, the product Y requires 5 hours in department A and 8 hours in department B. The constraints representing the above data:

a) $4x+5y \leq 1120$ and $7x+8y \leq 1400$	b) $4x+7y \leq 1400$ and $5x+8y \leq 1120$
c) $7x+5y \leq 1400$ and $8x+4y \leq 1120$	d) $7x+8y \leq 1120$ and $4x+5y \leq 1400$

58) A factory produces two products x and y. Each product passes through two departments A and B which have a capacity of 1120 hours and 1400 hours The Product X requires 7 hours in department A and 4 hours in department B, the product Y requires 8 hours in department A and 5 hours in department B. The constraints representing the above data:

a) $7x+4y \leq 1120$ and $8x+5y \leq 1400$	b) $7x+8y \leq 1400$ and $4x+5y \leq 1120$
c) $7x+5y \leq 1400$ and $8x+4y \leq 1120$	**d) $7x+8y \leq 1120$ and $4x+5y \leq 1400$**

59) A factory produces two products x and y. Each product passes through two departments M and N which have a capacity of 1120 hours and 1400 hours The Product A requires 7 hours in department M and 4 hours in department N, the product B requires 8 hours in department M and 5 hours in department N. The constraints representing the above data:

a) $7A+4B \leq 1120$ and $8A+5B \leq 1400$	b) $7A+8B \leq 1400$ and $4A+5B \leq 1120$
c) $7A+5B \leq 1400$ and $8A+4B \leq 1120$	**d) $7A+8B \leq 1120$ and $4A+5B \leq 1400$**

60) A factory produces two products x and y. Each product passes through two departments A and B which have a capacity of 1,100 hours and 1,400 hours The Product X requires 4 hours in department A and 5 hours in department B, the product Y requires 7 hours in department A and 8 hours in department B. The profit margin on x and y is Rs. 500 and Rs. 700 respectively. Objective function related to the given situation will be:

a) 1100x+1400y	**b) 500x+700y**
c) 600x+700y	d) 660,000x+980,000y

61) A company makes and sells two products X and Y. The contribution per unit is Rs. 250 for product X and 375 for product Y. due to various constraints, the company cannot make more than 750 units X and 500 units of Y in a month.
If x represents number of units of product X and y represents the number of units of product Y and C represents contribution, then relationship which represents maximum contribution would be C=250x+375y

a) True	b) False

62) A company makes and sells two products X and Y. the related information is as follows:

	X	Y
Contribution per unit (Rs.)	450	375
Maximum sales demand per month	2,800	1,200
Direct labour hour per unit	2	5
Machine hours per unit	6	7

A total of 10,000 direct labour hours and 22,000 machine hours are available per month. **The objective function (Z) and set of constraints represents the above situation would be:**

$Z=450x+375y$ $\quad 2x+5y \leq 22,000$ $\quad 6x+7y \leq 10,000$
$x \leq 2,800$ $\quad y \leq 1,200$ $\quad x, y \geq 0$

a) True	**b) False**

63) Which of the following constraints are represented by the given graph?

a) $2x+3y \leq 24$, $3x+5y \leq 45$, $x \leq 4$, $x,y \geq 0$	b) $2x+3y \leq 24$, $3x+5y \leq 45$, $y \leq 4$, $x \leq 0$, $y \geq 0$
c) $3x+2y \leq 24$, $5x+3y \geq 45$, $x \leq 4$, $y \geq 0$, $x \geq 0$	d) $3x+2y \leq 24$, $5x+3y \geq 45$, $y \leq 4$, $x \leq 0$ $x,y \geq 0$,

64) A company owns a machine which runs for 208 hours a month. The machine is used to make two parts X and Y. each part X takes 1 hour of machine time and each part Y takes 2 hours of machine time. If x represents the number of part X made in a month and y represents the number of part Y made in a month, which of the following statements/ inequalities is correct?

a) The company could make any quantity of X and Y but the total machine hours in a month cannot exceed 208.	b) $x+2y<208$ represents the boundary of maximum production in a month
c) $y \leq 208$ if $x=0$, represents the maximum production of Y in a month	d) both (a) and (b)

65) If $3x+7 \geq x+5 \geq 5x - 3$, then the inequality holds when x lies in the range

a) $7 \leq x \leq -3$	b) $5 \geq x \geq 7$
c) $3 \leq x \leq 5$	**d) $2 \geq x \geq -1$**

66) If $3x+7 \leq 5x - 3$, then the inequality holds when x lies in the range

a) Less than or equal to 7	**b) Greater than or equal to 5**
c) Less than or equal to 5	d) Greater than or equal to 2

67) A company makes and sells two products X and Y. the contribution per unit is Rs 250 for product X and 375 for product Y. due to various constraints, the company cannot make more than 750 units of X and 500 units of Y in a month. If x represents the number of product X, y represents the number of product Y and C represents contribution, which of the following relationship represents maximum contribution?

a) C=250x+375y	b) C=750x+500y
c) C=500x+125y	d) C=3x+1.33y

68) A company makes and sells two products X and Y. the related information is as follows:

	X	Y
Contribution per unit (Rs.)	450	375

Maximum sales demand per month	2,800	1,200
Direct labour hour per unit	2	5
Machine hours per unit	6	7

A total of 10,000 direct labour hours and 22,000 machine hours are available per month. Which of the following objective function (Z) and set of constraints represents the above situation?

a) Z=450x+375y x ≥ 2,800 6x+7y ≤ 10,000 2x+5y ≤ 22,000 y ≤ 1,200	**b) Z=450x+375y** **x ≤ 2,800** **2x+5y ≤ 10,000** **6x+7y ≤ 22,000** **y ≤ 1,200**
c) Z=450x+375y x ≤ 2,800 6x+7y ≤ 22,000 2x+5y ≤ 10,000 y ≥ 1,200	d) Z=450x+375y x ≥ 2,800 6x+7y ≤ 10,000 2x+5y ≥ 22,000 y ≤ 1,200

69) Tooba furniture (TF) manufacture chairs (C) and tables (T). the sale price and cost of production of the two articles are as follows:

Article	**Sale price**	**Cost of Production**
	Amount in Rs.	
C	850	700
T	1,300	1,200

Manufacturing time is 3 and 4 labour hours for each chair and table respectively. TF has a labour force of 20 workers and each worker can work a maximum of 180 hours. The total funds available for manufacturing are Rs. 4 million. The objective function (Z) and the constraints representing the given scenario are: (02)

a) Z=100C+150T 700x+1200y ≤ 4,000,000 3x+4y ≤ 3,600	**b) Z=150C+100T** **700x+1200y ≤ 4,000,000** **3x+4y ≤ 3,600**
c) Z=700C+1,200T 700x+1200y ≤ 4,000,000 3x+4y ≤ 180	d) Z=850C+1,300T 700x+1200y ≤ 4,000,000 3x+4y ≤ 3,600

70) In a business A and B invested the same amount @ 8% and 10% respectively. After 4 years B withdrew his investment. A keep his money for more 4 years. What is the extra amount that A gets as compare to B.

a)	b)
c)	d)

71) Hashim invested Rs. 20000 today and will withdraw amount after two years. What Will be the amount if he withdraws after 3 years at 12% compounded annually?

a) Rs. 25,088	**b) Rs. 28,098.56**	c) Rs. 47,488	d) Rs. 67,488

72) If Ali invested Rs. 30,000 today compounded annually for five years. There is no penalty if he withdraws money after 2 years. If he will receive Rs. 36,300 after 2 years

1) What is the interest rate?
2) What money he gets, if he withdraws after 3 years

a) 10%, Rs.36300	b) 10.5%, Rs.39450	**c)10%, Rs. 39,930**	None of these

73) A person invested Rs. 100,000 at the start of each year for 6 years @8% simple interest for first 3 years and 10% simple interest for last 3 years. Find the total amount he received at the end of 6 years.

a) Rs 698,000	b) Rs 654,000
c) Rs 798,000	d) None of these

74) Basit wants to took a loan for 5 years Which would be the least beneficial for him?

a) 11 % simple interest	b) 8% compounded quarterly
c) 10% compounded annually	d) 9% compounded semi annually

75) A person invested some amount today @1.8% per quarter for 10 years find his investment if he receives Rs. 10 million.

a) 8.356 million	b) 4.13 million	**c) 4.899 million**	d) None of these

76) Basit took Rs. 200,000 from bank at 13.5% rate per year. What is the total amount after 5 years?

a) 356,700	**b) 376,712**	c) 335,000	d) 337,500

77) A person took a loan of Rs. 120,000 at simple interest per annum. There is no penalty on withdrawing after 2 years. However, the total amount he receives after 4 years is Rs. 156,000. Which of the following statement is/are correct?

a) If he withdraws after 3 years, he would receive a total amount of Rs. 138000	**b) The simple interest rate is 7.5%**
c) both a & b	d) None of these

78) Mr. Ikram invests Rs. 400,000 at the start of each quarter@12% for 8 years. What will be the total amount after 8 years?

a) Rs. 5,510,263	b) Rs. 4,919,877
c) Rs. 21,101,103	**d) Rs. 21,631,137**

79) A person has enough amount to invest for 9 years, if Rs. 1 million is received after 9 years (end of year) at the rate of 7%, What is amount he has to invest each year (at the start)?

a) Rs.83,486	b) Rs.78,025	c) Rs.543,933	d) Rs.613,497

80) If the fees earned by school in 10th year is 5 million and fee is increased each year by 10% and number of students increase by 5%. What was the fee earned by school in 1st year?

a) Rs.3.07Million	b) 1.93 million	**c) 1.37 million**	d) 0.81 million

81) If 5 million is invested for 3 years and yield 6 million at compounded quarterly interest rate. If this amount is not withdrawn and further invested at the same interest rate, what will be the fee earned by school in further 3 years?

a) Rs.7.2M	b) Rs.6M	c) Rs.7.5M	d) Difficult to calculate

82) If Rs. 500,000 is invested in a scheme @ 1.5% per month simple interest for 5 years then:

a) He will get gain of Rs. 150,000 per annum	b) He will get gain of Rs. 75,000 per annum
c) **He will get gain of Rs. 450,000 after 5 years**	d) All of these are correct.

83) Majid borrowed Rs. 600,000 at some simple interest x for 2 years and 3 months and paid 39,960 in excess, find interest rate?

a) 4.23%	**b) 2.96%**	c) 29.6%	d) 42.3%

84) Mr. Ali invested Rs. 10,000 in 6 years, in first 3 years he received return at 8% and next 3 years he received return at 10%. What will be total amount he will receive after 6 years?

a) Rs. 16,766.8	b) Rs. 15,400

c) Rs. 16,120	d) Rs. 10,000

85) Sohail is receiving interest from Fast Bank Limited (FBL) at 14% compounded semi-annually. Slow bank limited (SBL) has introduced a scheme whereby interest would be compounded on a quarterly basis.
SBL should offer a minimum interest rate of 13.16% to motivate Sohail to shift his investment from FBL to SBL is:

a) True	**b) False**

86) Nasir had taken Rs. 900,000 from his office at 13.5% simple interest for a period of 5 years and 5 months. The principal amount was paid at the expiry of loan period.
He paid interest of Rs. 595,000 during the period of loan.

a) True	**b) False**

87) Mr. A invested Rs. 10,000 @4% compounded quarterly and received the total amount at the end of 5th year. Find the amount.

a) Rs 12,000	b) Rs 13,202
c) Rs 12,202	d) Rs 15,302

88) Government has issued a five years bond of Rs. 200,000. On maturity the buyer will get Rs 300,000. If the current interest rate is 8% per annum, is purchasing the bond worth?

a) Yes, as present value of Rs 300,000 is more than Rs 200,000	b) No, as present value of Rs 300,000 is more than Rs 200,000
c) Yes, as present value of Rs 300,000 is less than Rs 200,000	d) No, as future value of Rs 200,000 is more than Rs 300,000

89) Government has issued a five years bond of Rs. 200,000. On maturity the buyer will get Rs 300,000. If the current interest rate is 9% per annum, is purchasing the bond worth?

a) Yes, as Future Value of Rs 200,000 at current interest rate is more than Rs 300,000	b) No, as Future value of Rs 200,000 at current interest rate is more than Rs 300,000
c) Yes, as present value of Rs 300,000 at current interest rate is more than Rs 200,000	**d) No, as present value of Rs 300,000 at current interest rate is less than Rs 200,000**

90) Find the value invested now, if he received 8,000 at the end of each year from year 4 to year 12 at the rate of11% compounded annually?

a) 51939	**b) 32,389**
c) 27119	d) 57652

91) Ali invests Rs. 2 million and get 0.5 million at redemption. If n = 3, which of the following interest rate is best?

a) 7.15% monthly	b) 7.23% bi-monthly
c) 8.1% simple interest	**d) 7.51% quarterly**

92) Alia borrowed Rs. 700,000 from Sara for a period of 2 years and 3 months at r% simple interest. She paid a total of 950,000 at the end of loan period. the value of "r" is

a) 14.29%	b) 15.53%
c) 15.87%	d) 17.86%

93) Suman borrowed Rs. 900,000 at simple interest of 7.5% per annum. At the end of the loan period she repaid a total of 1,372,500. Period of the loan was:

a) 7 years	b) 7 years 3 months
c) 7 years 6 months	d) 8 years

94) Haroon has borrowed a certain amount at an interest of 12% compounded semi-annually. In how many years the amount owed would double?

a) 3 years	b) 5 years
c) 6 years	d) 8 years

95) Hashir has borrowed "x" amount at an interest of 13% compounded semi-annually. If he does not return the amount, in which of the following years the amount owed would be 2x?

a) Third year	b) Fifth year
c) sixth years	d) eighth years

96) Rida has borrowed a certain amount at an interest of 9% compounded quarterly. In which year the amount owed would be double?

a) 6^{th} year	b) 7^{th} year
c) 8^{th} year	d) 9^{th} year

97) Saad borrowed Rs 600,000 from Fahad for a period of 3 years and 9 months at r% simple interest. He paid Rs 400,000 in excess of the borrowed amount at the end of loan period. The value of "r" is:

a) 22.22%	**b) 17.78%**
c) 17.09%	d) 16.67%

98) Saad borrowed Rs 90,000 from Fahad for a period of 5 years and 10 months at r% simple interest. He paid Rs 40,000 in excess of the borrowed amount at the end of loan period. The value of "r" is:

a) 12.8%	b) 8.71%
c) 5.57%	**d) 7.61%**

99) If Present value is 1,000,000 and interest is 750,000 in 4 years and 10 months. Find the simple interest rate?

a) 15.51%	b) 5.17%
c) 7.75%	d) None of these

100) Mani borrowed Rs 900,000 from Rani for a period of 4 years and 10 months on simple interest. He paid Rs 750,000 in excess of the borrowed amount at the end of loan period. The rate of interest is:

a) 18.33%	**b) 17.24%**
c) 20.33%	d) 16.67%

101) Hanif borrowed Rs 800,000 from Rehan for a period of 4 years and 5 months on simple interest. He paid Rs 650,000 in excess of the borrowed amount at the end of loan period. The rate of interest is:

a) 18.06%	b) 18.13%
c) 18.40%	d) 20.31%

102) Baber borrowed Rs. 900,000 at simple interest of 9.68% per annum. At the end of the loan period he repaid a total of Rs. 1,510,000. Period of the loan was"

a) 5 years and 9 months	b) 6 years
c) 6 years and 3 months	**d) 7 years**

103) Baber borrowed Rs. 900,000 at simple interest of 8.34% per annum. At the end of the loan period he repaid a total of Rs. 1,500,000. Period of the loan was"

a) 7 years	b) 7 years and 3 months
c) 7 years and 9 months	**d) 8 years**

104) Faraz borrowed Rs. 1,000,000 at simple interest of 8.5% per annum. At the end of the loan period he repaid a total of Rs. 1,510,000. Period of the loan was"

a) 5 years and 9 months	**b) 6 years**
c) 6 years and 3 months	d) 7 years

105) Saleem borrowed Rs. 500,000 from a bank at simple interest of 2% per month for a period of 3 years. the principal is payable in equal monthly instalments, along with interest. Which of the following statements is correct?

a) His monthly instalments would be Rs. 23,000	b) He would pay Rs. 100,000 per annum in interest
c) He would have paid an additional amount of Rs 300,000 by the end of 3 years	**d) At the end of year 1, his balance principal amount would be Rs. 333,333.33**

106) Tom and Jerry both invested same amount for 8 years. If Tom's rate of return is 9% and Jerry's rate is 10%compounded annually then how much more percent amount Jerry will have after 8 years than Tom?

a) 1%	**b) 8%**
c) 15.1%	d) None of these

107) An investment of 1.5 million is made in a business for 4 years and gain is 0.5 million. Find rate of gain undercompound interest?

a) 6.54%	b) 8.5%
c) 7.45%	d) 9.4%

108) John and Johnson have invested same amounts in two different investment schemes. John is getting a return of 9% compounded annually whereas Johnson gets a return of 10% compounded annually, the amount of Johnson' s interest over a period of six years would exceed the amount of John's interest by

a. 9.44%	b. 5.63%	c. 31.47%	**d. 13.95%**

109) Naeem and Karim have invested same amounts in two different investment schemes. Naeem is getting a return of 7% compounded annually whereas Karim gets a return of 9% compounded annually, the amount of Karim's interest over a period of five years would exceed the amount of Naeem's interest by:

a) 25.26%	b) 9.70%

c) 13.60%	**d) 33.80%**

110) Saeed invested Rs 400,000 for ten years after which he received a lump sum amount of Rs 900,000. If he earned 8.50% interest compounded quarterly during last five years which of the following rate compounded quarterly did he earn during first five years

a) 7.89%	b) 8.25%
c) 8.39%	d) 8.50%

Answer

$$900{,}000 = 400{,}000\left(1+\frac{r}{4}\right)^{4\times5}\left(1+\frac{0.085}{4}\right)^{4\times5}$$

By solving on calculator we get r=0.078843=7.89%

111) Anees invested Rs 500,000 for six years after which he received a lump sum amount of Rs 822,531. If he earned 10% interest compounded annually during last four years, which of the following rate compounded quarterly did he earn during first two years

a) 5.87%	b) 8.25%
c) 8.39%	d) 8.50%

Answer

$$822{,}531 = 500{,}000\left(1+\frac{r}{4}\right)^{4\times2}(1+0.10)^{4}$$

By solving on calculator we get r=0.05869515029 =5.87%

112) Ali invested Rs 500,000 for5 years after which he received a lump sum amount of Rs 762,150. If he earned 10% interest compounded annually during last 2 years, what rate of interest compounded annually did he earn during the first three years?

a) 6%	b) 7%
c) 8%	d) 9%

Answer $762{,}150 = 500{,}000(1+r)^{3}(1+0.10)^{2}$

By solving on calculator we get r = 0.0800114499=8%

113) Jamil invested Rs. 6 million with a real estate firm. The firm returned him Rs. 8 million after four years. The effective annual rate of interest on his investment was:

a) 7.46%	b) 7.00%
c) 8.25%	d) 8.33%

1. A person wants to invest some money for 6 years. He is offered four different interests. Which one is better?

a) 15% simple interest	b) 10% compounded annually
c) 9% compounded semi-annually	d) 8% compounded monthly

114) Ali and Kashif invested equal sum in two different banks at 9% and 10% interest compounded annually. In 6years how much percent of sum will Kashif earn more than Ali?

a) **9.45%**	**b)** 8.45%	c) 7.45%	d) 6%

115) If Naina and Maina both invested same amount for 6 years and the rate of interest are 9% and 10% compounded annually respectively. Then how much more percent amount Maina will have after 6 years than Naina?

a) 7.63%	b) 8%
c) 9.45%	**d) None of these**

116) Kareem and Jaffer have invested same amounts in two different investment schemes. Kareem is getting areturn of 8% compounded annually whereas Jaffer gets a return of 10% compounded annually, the amount ofJaffer's interest over a period of six years would exceed the amount of Kareem's interest by

a) 11.25%	b) 12%	c) **31.47%**	d) 8%

117) Mr. A invested 200,000 in an account today. He also deposits 20,000 quarterly in this account and made first payment today. If the interest is 8% compounded quarterly. What will be total value of investment after 5 years?

a) Rs 992,855	b) Rs 222,855
c) Rs 792,855	d) Rs 692,855

6 Discounted Cash flows

118) A person withdraws Rs. 300,000 per annum from 4th year to 10th year at 7% compounded annually, how much should he invest now?

a) Rs. 1,319,780	b) Rs 1,233,439	c) Rs 1,090,911	d) None of these

119) A bank launched a new scheme where it offers Rs. 400,000 per year for indefinite time after 8 years. How much amount should bank require to collect from customer today if it offers 5% interest compounded annually?

a)8 million	**b)5.69 million**	c.5.41 million	d)5.6 million

120) Calculate the IRR of the project

Year	Cash flows
0	(1,500,000)
1	100,000
2	200,000
3	1,450,000

a) 0.573%	b) .573%	**c) 5.73%**	d) 57.3%

121) Calculate IRR of the project:

Year	Cash flows
0	(5,000,000)
1	1,000,000
2	5,200,000
3	850,000

a. **19.2%**	b. 12.3%	c. 0.192%	d. 14.4%

122) A project costing 4 million is expected to yield Rs. 600,000, Rs. 800,000 & Rs. 4,000,000 at the end of each year for next 3 years respectively. The IRR of the project is?

a. 12.22%	b. 21.22%	c. 22.12%	d. 22.21%

123) A project-costing 2 million is expected to yield Rs. 100,000, Rs. 200,000 & Rs. 2,300,000 at the end of each year for next 3 years respectively. The IRR of the project is?

a) 8.8%	**b) 9.7%**	c) 9.2%	d) 11.51%

124) A project has a net present value of Rs. 26,708 n discount rate is 13%. At what discount rate will the net present value of project be Rs. 35,612?

a) 15%	b) 17%	c) 19%	**d) 11%**

125) A project has a NPV of Rs. 6,172 when discount rate is 9%. What will be the net present value if discount rate is increased to 12%

a) 9,438	b) 10,292	**c) 3,291**	d) 9,163

126) The project has an NPV of Rs. 2,210 at a discount rate of 11% & (1,918) at a discount rate of 15%. Estimate the IRR of the project.

a) 13.14%	b) 17.14%	c) 16.85%	d) 12.85%

127) A project costing 3 million is expected to yield Rs. 800,000, Rs. 1,200,000 & Rs. 1,800,000 at the end of each year for next 3 respectively. The IRR of the project is?

a) 18.11%	b) **11.18%**	c) 12.22%	d) 22.21%

128) The project has an NPV of Rs. 5641 at a discount rate of 8% & (2354) at a discount rate of 10%. Estimate the IRR of the project:

a) 9.41%	b) 11.43%	c) 38.89%	d) 20.70%

129) Two companies made profits from investments in different projects:

Year	1	2	3	4
Company A	600,000	900,000	1,200,000	-
Company B	600,000	750,000	850,000	900,000

Find the rate at which NPV of both companies will be same?

e) 14.47%	f) 14.60%	**g) 54.65%**	h) 15.2%

130) You are given some annuity due of Rs. 200,000, find the sum of ordinary annuity if r= 12% compounded quarterly?

a) Rs. 206,000	b) **Rs. 194,174.76**

c) Rs. 200,000	d) None of these

131) A research foundation was established by a fund of Rs. 500,000 invested at rate that would provide 20,000 payments at the end of each year for forever. What interest rate was being earned on fund?

a) 5%	b) 3%	**c) 4%**	d) 6%

132) If Rs 1000 is invested yearly for 3 years @7% compounded annually and some other unknown amount is invested annually for next 2 years @5% compounded annually and Rs 500,000 is received at the end of five years, then find the amount invested in last two years.

a) Rs 212,173.45	**b) Rs 242,173.45**
c) Rs 202,173.45	d) Rs 302,173.45

133) Zafar has purchased a motor cycle worth Rs. 40,000 from his friend who has given him the following payment options.

a) pay Rs. 52,000 at the end of four years.
b) pay Rs. 12,000 annually at the end of next 4 years
c) pay the full amount now

Zain's cost of funds is 10 %

Match the following

Most beneficial	**Option (A)**
second most beneficial	**Option (B)**
third most beneficial	**Option (C)**

134) Rashid wants to obtain a bank loan. Bank a offers a nominal rate of 14% compounded monthly, Bank B offers a nominal rate of 14.5% compounded quarterly; and Bank c offers an effective rate of 14.75%

Arrange the given banks in terms of charging lower interest using drag and drop option

Bank A	**Sequence 1**
Bank B	**Sequence 2**
Bank C	**Sequence 3**

135) Saleem is planning to invest in a scheme whereby he would be required to invest Rs. 160,000 annually (at the start of the year) for 5 years. If interest rate is 14% compounded annually, he would receive Rs. 1,057,617 at the end of 5^{th} year.

a) True	b) False

136) To derease present value of a project, the discount rate should be adjusted increased

137) A company intends to invest Rs 6 million into a project which would yield 10,12 and 14 percent during first three years respectively. The company would also recover the original investment after 3 years. If the company's cost of capital is 9%, NPV of the project would be:

a) Rs 393,745	**b) Rs 438,204**
c) Rs 648,634	d) Rs 1,805,103

138) A company intends to invest Rs 9 million into a project which would yield 8, 12 and 14 percent during first three years respectively. The company would also recover the original investment after 3 years. If the company's cost of capital is 9%, NPV of the project would be:

a) Rs 393,745	b) Rs 438,204
c) Rs 648,634	**d) Rs 492,167**

139) A company intends to invest Rs 4 million into a project which would yield 10, 12 and 13 percent during first three years respectively. The company would also recover the original investment after 3 years. If the company's cost of capital is 10%, NPV of the project would be:

a) Rs 393,745	b) Rs 438,204
c) Rs 156,273	d) Rs 1,805,103

140) A company intends to invest Rs 4 million into a project which would yield 10, 12 and 13 percent during first three years respectively. The company would also recover the original investment after 3 years. If the company's cost of capital is 9%, NPV of the project would be:

a) Rs 261,248	b) Rs 438,204
c) Rs 648,634	d) Rs 1,805,103

141) A company intends to invest Rs 4 million into a project which would yield 10, 12 and 14 percent during first three years respectively. The company would also recover the original investment after 3 years. If the company's cost of capital is 12%, NPV of the project would be:

a) Rs 393,745	b) Rs 438,204

c) Rs 648,634	**d) Rs (14,486)**

142) Iram invested Rs.2 million and received Rs.2.5m after 3 years. Calculate the rate of interest?

a) 7.51% compounded monthly	**b) 7.51% compounded quarterly**
c) 7.51% compounded bi-monthly	d) 7.51% compounded annually

143) A company intends to invest Rs 5 million into a project which would yield 10, 12 and 14 percent during first three years respectively. The company would also recover the original investment after 3 years. If the company's cost of capital is 10%, NPV of the project would be:

a) Rs 393,745	**b) Rs 232,908**
c) Rs 648,634	d) Rs 1,805,103

144) A company intends to invest Rs 5 million into a project which would yield 10, 12 and 16 percent during first three years respectively. The company would also recover the original investment after 3 years. If the company's cost of capital is 10%, NPV of the project would be:

a) Rs 393,745	b) Rs 438,204
c) Rs 308,039	d) Rs 1,805,103

145) A project costing Rs. 2 million is expected to yield Rs. 300,000, Rs. 400,000, Rs. 1,900,000 at the end of each of the next 3 years respectively. The IRR of the project is:

a) 11.51%	**b) 10.66%**
c) 11.15%	d) 10.18%

146) A project costing Rs. 2.5 million is expected to generate cash flows of Rs. 200,000, Rs. 300,000, Rs. 2,900,000 at the end of each of the next three years respectively. The IRR of the project is 11.7%

a) True	b) False

147) If the interest rate is 9% compounded monthly, the value of perpetuity of Rs. 3,000 per month would be 400000.

148) Rashid has savings of 3.6 million whereas sajjad has savings of 5.4 million. If Rashid invest his savings at 9% compounded quarterly whereas Sajjad invest his savings at 7% compounded annually, in how many years would the value of Rashid's savings exceed that of Sajjad's savings.

a) 9 years	b) 14 years
c) 19 years	d) 24 years

149) Miss Salma requires a sum of Rs. 300,000 after three years from now and Rs. 500,000 after 5 years from now, for the purpose of education of her son. She is planning to deposit quarterly amounts in a bank account from which she would draw the desired amounts at the required time. If interest rate is 12% compounded Quarterly, which of the following amounts should Salma deposit at the start of each quarter?

a) Rs 76,766	b) Rs 74,530
c) Rs 31,797	d) Rs 32,751

150) A certain sum of money lent out at simple interest amount to 690 in three years and 750 in five years. The sum lent is:

a) Rs 500	**b) Rs 600**
c) Rs 700	d) Rs 800

151) If Rs 100,000 is invested yearly for 3 years @7% compounded annually and some other unknown amount is invested annually for next 2 years @5% compounded annually and Rs 500,000 is received at the end of five years, then find the amount invested in last two years.

a) Rs 71,003.55	b) Rs 60,000
c) Rs 70,000	d) Rs 80,000

152) Which of the following statements is CORRECT?

a) **It is impossible to find the future value of a perpetuity**.	b) It is impossible to find the present value of a perpetuity
c) Perpetuity factor is square of the cost of capital	d) In investment appraisal, an annuity might be assumed when a constant annual cash flow is expected for a long term into the future.

153) A bank launched a new scheme where it offers 400,000 per year for indefinite time after 8 years. How much amount should bank require to collect from customer today if it offers 5% interest compounded annually?

a) Rs. 8,000,000	b) Rs. 5,414,714
c) Rs. 5,685,451	d) None of these

154) A bank is planning to offer a unique product to its customers whereby it would pay Rs 250,000 per annum for an indefinite period commencing from the end of year 6. How much amount should the bank ask its customers to pay now, if the rate of interest that the bank can pay, is 5% compounded annually?

a) Rs 3,191,221	b) Rs 3,547,829
c) Rs 3,917,631	d) Rs 3,960,498

155) Asif plans to invest Rs 5,000 every year starting from today for next 3 years. Interest rate is 10% per annum compounded annually. Future value of the annuity is:

a) Rs 16,500	b) Rs 17,050
c) Rs 17,600	**d) Rs 18,205**

156) Sami plans to invest Rs 8,000 every year for 3 years starting from today. Interest rate is 10% per annum compounded semi-annually. At the end of year 3 he will receive:

a) Rs 29,128	**b) Rs 29,265**
c) Rs 26,480	d) Rs 27,871

157) Kaleem plans to invest Rs 8,000 every year for 3 years starting from today. Interest rate is 10% per annum compounded annually. At the end of year 3 he will receive:

a) Rs 26,480	b) Rs 28,328
c) Rs 29,128	d) Rs 31,944

158) Razia plans to invest Rs 9,000 every year for 3 years starting from today. Interest rate is 10% per annum compounded annually. At the end of year 3 she will receive:

a) Rs 29,790	b) Rs 31,869
c) Rs 35,937	**d) Rs 32,769**

159) Zaki plans to invest Rs 6,000 every year for 3 years starting from today. Interest rate is 10% per annum compounded annually. At the end of year 3 he will receive:

a) Rs 23,958	**b) Rs 21,846**
c) Rs 21,246	d) Rs. 19,860

160) Zaki plans to invest Rs 6,000 every year for 4 years starting from today. Interest rate is 10% per annum compounded annually. At the end of year 4 he will receive:

a) Rs 30,000	**b) Rs 30,631**
c) Rs 33,600	d) Rs. 26,400

161) Razia plans to invest Rs 9,000 every year for 4 years starting from today. Interest rate is 10% per annum compounded annually. At the end of year 4 she will receive:

a) Rs 29,790	b) Rs 31,869
c) Rs 35,937	**d) Rs 45,945.9**

162) Which of the following statement is CORRECT?

a) Discounting estimates the present-day equivalent of a future cash flow at a specified time in the future at a given rate of interest
b) Multiplying by a discount factor is the same as multiplying by a compounding factor
c) The present value of a cash flow is the recipient of its future value
d) Present value fails to appraise large projects with multiple cash flows.

163) Raza wants to save money over a period of ten years in order to meet the expenses to be incurred on higher education of his son. He has recently invested a sum of Rs 200,000 and plans to further invest Rs 20,000 at the end of each quarter, which of the following amount will be available to him at the end of 10th year if he earns a profit of 6% per annum compounded quarterly?

a) Rs 1,448,161.56	b) Rs 1,321,027.61
c) Rs 992,497.74	d) Rs 718,018.61

164) Zahid wants to save money over a period of ten years in order to meet the expenses to be incurred on higher education of his son. He has recently invested a sum of Rs 200,000 and plans to further invest Rs 20,000 at the end of each quarter, which of the following amount will be available to him at the end of 10th year if he earns a profit of 6% per annum compounded quarterly?

a) Rs 1,448,161.56	b) Rs 1,321,027.61
c) Rs 992,497.74	d) Rs 718,018.61

165) Aman wants to save money over a period of eight years in order to meet the expenses to be incurred on higher education of his son. He has recently invested a sum of Rs 100,000 and plans to further invest Rs 20,000 at the end of each quarter, which of the following amount will be available to him at the end of 8th year if he earns a profit of 8% per annum compounded quarterly?

a) Rs 1,234,876.08	b) Rs 1,117,221.68
c) Rs 1,072,994.65	d) Rs 623,788.75

166) A bank launched a new scheme where it offers Rs. 8000 per year for indefinite time after 8 years. How much amount should bank require to collect from customer today if it offers 8% interest compounded annually?

a) Rs 54,027	b) Rs 100,000	**c)** Rs 63,017	**d) Rs 58,349**

167) If the discount rate is 10%, the present value of Rs X received at the end of each year for the next five years is equal to:

a) 3.5X	b) 4.17X
c) **3.79X**	d) 5X

168) If the discount rate is 11%, the present value of Rs X received at the end of each year for the next five years is equal to:

a) 3.17X	b) 4.10X
c) 3.7X	d) 5X

169) If the discount rate is 8%, the present value of Rs X received at the end of each year for the next four years is equal to:

a) 2.2X	**b) 3.31X**
c) 1.2X	d) None of these

170) If the discount rate is 12%, the present value of Rs X received at the end of each year for the next four years is equal to:

a) 3.4X	**b) 3.04X**
c) 4X	d) 3.6X

171) If the discount rate is 12%, the present value of Rs X received at the end of each year for the next five years is equal to:

a) 6X	b) 5X
c) 3.6X	d) 4.03X

172) If the discount rate is 9%, the present value of Rs X received at the end of each year for the next four years is equal to:

a) 3.24X	b) 3.42X
c) 3.89X	d) 4X

173) If the discount rate is 14%, the present value of Rs X received at the end of each year for the next five years is equal to:

a) 3.91X	**b) 3.43X**
c) 3.7X	d) 5X

174) Ashfaq is planning to invest in a scheme whereby he would be required to invest Rs 130,000 annually (at the start of the year) for 5 years. If the interest rate is 13% compounded annually, what amount would he receive at the end of the 5th year?

a) Rs 842,435	**b) Rs 951,952**
c) Rs 964,952	d) Rs 1,075,706

175) Abid is planning to invest in a scheme whereby he would be required to invest Rs 120,000 annually (at the start of the year) for 6 years. If the interest rate is 8% compounded annually, what amount would he receive at the end of the 6th year?

a) Rs 648,500	b) Rs 777,600
c) Rs 880,311	**d) Rs 950,736**

176) The rate of interest is 8% per annum compounded monthly, the value of perpetuity of Rs 2,500 per month would be:

a) Rs 375,000	b) Rs 187,500
c) Rs 37,500	d) Rs 31,250

177) If the rate of interest is 12% per annum compounded monthly, the value of perpetuity of Rs 2,500 per month would be achieved by investing:

a) Rs 250,000	b) Rs 20,833
c) Rs 125,000	d) Rs 187,500

178) If the rate of interest is 6% per annum compounded monthly, the value of perpetuity of Rs 2,500 per month would be:

a) Rs 41,667	b) Rs 166,667
c) Rs 250,000	**d) Rs 500,000**

179) If the rate of interest is 8% per annum compounded monthly, the value of perpetuity of Rs 2500 per month would be

a) 31,250	b) 37,500
c) 187,500	**d) 375,000**

180) If the rate of interest is 8% per annum compounded quarterly, the value of perpetuity of Rs 3,500 per quarter would be:

a) Rs 131,250	**b) Rs 175,000**
c) Rs 262,500	d) Rs 525,000

181) If the rate of interest is 5% per annum compounded quarterly, the value of perpetuity of Rs 2,500 per quarter would be achieved by investing

a) Rs 50,000	**b) Rs 200,000**
c) Rs 250,000	d) Rs 500,000

182) If the rate of interest is 15% per annum compounded monthly, the value of perpetuity of Rs 1,975 per month would be achieved by investing

a) Rs 375,000	b) Rs 187,500
c) Rs 158,000	d) Rs 37,500

183) If the rate of interest is 10% per annum compounded monthly, the value of perpetuity of Rs 3,000 per month would be achieved by investing

a) Rs 400,000	b) Rs 300,000
c) Rs 270,000	**d) Rs 360,000**

184) Which of the following is correct?

a) Net Present Value (NPV) is a financial metric that seeks to capture the total value of a potential investment opportunity
b) Multiplying by a discount factor is the same as multiplying a compounding factor.
c) The present value of a cash flow is the reciprocal of its future value
d) Present value fails to appraise large projects with multiple cash flows

185) A company intends to invest Rs 4 million into a project which would yield 12,14 and 16 percent during three years respectively. The company would also recover the original investment after 3 years. Company's cost of capital is 10%, NPV of the project would be:

a) 385,274	b) 436,364
c) 480,841	d) 1,380,015

186) A company intends to invest Rs 3 million into a project which would yield 10,12 and 14 percent during three years respectively. The company would also recover the original investment after 3 years. If the company's cost of capital is 10%, the NPV of the project is:

a) Rs 139,745	b) Rs 45,582
c) Rs 1,046,582	d) Rs 3,139,745

187) A company intends to invest Rs 3 million into a project which would yield 10,12 and 14 percent during three years respectively. The company would also recover the original investment after 3 years. If the company's cost of capital is 8%, the NPV of the project is:

a) Rs 277,778	**b) Rs 301,326**
c) Rs 333,410	d) Rs 919,830

188) A company intends to invest Rs 3 million into a project which would yield 10,15 and 20 percent during first three years respectively. The company would also recover the original investment after 3 years. If the company's cost of capital is 12%, the NPV of the project would be:

a) Rs 1,350,000	b) Rs 168,754
c) Rs (39,783)	**d) Rs 189,003**

189) A project costing Rs. 2.5 million is expected to generate cash flows of Rs. 200,000, Rs. 300,000, Rs. 2,900,000 at the end of each of the next three years respectively. The IRR of the project is:

a) 9.9%	b) 10.4%
c) 11.7%	d) 12.8%

190) A project costing Rs. 2 million is expected to yield Rs. 300,000, Rs. 400,000, Rs. 1,900,000 at the end of each of the next 3 years respectively. The nearest IRR approximation of the project is:

a) 10.18%	**b) 10.66%**
c) 11.15%	d) 11.51%

191) A project costing Rs. 2 million is expected to yield Rs. 100,000, Rs. 200,000, Rs. 2,300,000 at the end of each of the next 3 years respectively. The nearest IRR approximation of the project is:

a) 8.8%	**b) 9.7%**
c) 9.2%	d) 11.51%

192) A project costing Rs. 4 million is expected to yield Rs. 600,000, Rs. 800,000, Rs. 4,000,000 at the end of each of the next 3 years respectively. The nearest IRR approximation of the project is:

a) 11.52%	b) 11.84%
c) 12.22%	d) 12.71%

193) Ali plans to invest Rs. 8,000 every year for 3 years starting from today. Interest rate is 10% per annum compounded annually. At the end of year 3 he will receive:

a)	Rs 26,480	b)	Rs 26,328
c)	**Rs 29,128**	d)	Rs 31,944

194) Project A would provide annual inflows of Rs. 525,000 Rs. 648,000, Rs 853,000 and Rs 2,844,000 at the end of year 1 to 4 respectively, whereas project B would yield annual inflows of Rs 947,000, Rs. 1,155,000 and Rs. 2,068,000 from year 1 to 3 respectively. The discount rate at which both projects would have same net present value is:

a)	**18.27%**	b)	18.83%
c)	19.31%	d)	19.73%

195) Project X would provide annual inflows of Rs. 650,000 Rs. 500,000, Rs 1,000,000 and Rs 2,500,000 at the end of year 1 to 4 respectively, whereas project Y would yield annual inflows of Rs 849,650, Rs 1,166,800 and Rs 2,068,000 from year 1 to 3 respectively. The discount rate at which both projects would have same net present value is:

a) 16.31%	**b) 17.51%**
c) 18.27%	d) 19.31%

7 ***Data-Collection and representation***

196) Which of the following statement is correct about Population Parameter?

(i) All the items are present in the data from which parameter is determined.

(ii) 50 people selected through random sampling can be used to find parameter.

a) Both statements are correct	b) Both statements are not correct
c) Only statement (i) is correct	d) Only statement (ii) is correct

197) Which of the following statements are correct regarding Class boundary?

1) There is no gap between the classes;
2) Midpoint of the upper-class limit of one class and the lower-class limit of the subsequent class.

a) Both statements are correct	b) Both statements are not correct
c) Only statement (i) is correct	d) Only statement (ii) is correct

198) Which of the following is/are correct regarding primary data?

1. Data collected by interviewing students in school.
2. Data collected by interviewing the parents of students.

a) Only statement (1) is correct	b) Only statement (2) is correct
c) **Both statements are correct**	d) Both statements are not correct

199) Which of the following statement is/are correct?

1. Discrete data has gaps between classes
2. Bar charts are used for continuous data

a) Both statements are correct	b) Both statements are not correct
c) Only statement (1) is correct	d) Only statement (2) is correct

200) Which of the following is correct?

1) The grouped frequency of discrete data does not contain gaps between the classes.
2) Discrete data has gap between them

a) Only statement (1) is correct	b) **Only statement (2) is correct**
c) Both statements are correct	d) Both statements are not correct

201) Which of the following is discrete data?

a) No. of children of an employee	b) Weight of students
c) Height of students	d) None of these

202) Construction of frequency distribution:

a) Helps in deleting the data	**b) Begins by recording the number of times a particular value occurs**
c) Is the only way to assess mode od population.	d) All of the above

203) Which of the following is correct?

a) Results of sampling enquiries or a census is called raw data	b) The data which is collected specifically for the ongoing investigation is called primary data
c) The data which is stored after classification is called secondary data	**d) Both (a) and (b)**

204) Data can be collected through which of the following method(s):

a) Direct Observation	b) National Census
c) CCTV recordings	**d) All of the above**

205) Which of the following statements is/are correct?

(i) A grouped frequency distribution of discrete data has gaps between the classes.

(ii) Discrete data can be converted into continuous data.

a) Only statement 1 is correct	b) Only statement 2 is correct
c) Both statements are correct	d) None of the above are correct

206) Which of the following statement is correct?

a) A grouped frequency distribution of discrete data has gaps between the classes	b) Discrete data can be converted into continuous data

c) In the case of discrete data, the mid-point of upper-class limit of one class and lower-class limit of subsequent class is termed as class boundary	**d) All of the above**

207) Which of the following statements is correct?

a) Bar charts are usually used for plotting continuous data	b) Bar charts are usually used for plotting discrete data
c) Bar charts can be plotted horizontally or vertically.	**d) Both (b) and (c)**

208) Which of the following statements regarding Bar charts is correct?

a) Bar charts are usually used for plotting continuous data	**b) Bar charts can be plotted horizontally or vertically.**
c) Bar charts are usually used for plotting vertical data only	d) Both (a) and (b)

209) Which of the following statements as regards Histogram is correct?

a) A vertical rectangle is drawn to represent each class of the frequency distribution	b) The frequency of the class is represented by the height of rectangles
c) Histogram cannot represent continuous data	d) Both (a) and (b)

210) Which of the following statements as regards Histogram is correct?

a) Histogram cannot represent continuous data	b) **A vertical rectangle is drawn to represent each class of the frequency distribution**
c) The frequency of the class is represented by the height of rectangles	d) Histogram are used to compare variables

211) A set of exam scores is represented by the following stem and leaf display:

4	5	6	8								
5	3	4	5	6	9						
6	2	3	5	6	6	9	9				
7	0	1	1	3	3	4	5	5	5	7	8
8	1	2	3	6	9						
9	3	5	7	8							

The mode for the data set is

a) 75	b) 71
c) 78	d) 69

212) 81, 81, 81, 81, 21, 21, 21, 21, 21, 90, 90, 90, 90. Find mode of the data?

a) 21 only	b) 21 & 81 only
c) 21 & 90 only	d) None of these

213) 21, 21, 21, 21, 21, 28, 28, 28, 28, 29, 29, 29, 29. Find mode of the data?

a) **21 only**	b) 21 & 28 only	c) 21 & 29 only	d) None of these

214) The following stem and leaf display show the number of pizza slices eaten by contestants in a recent pizza eating contest.

3	1					
4	0					
5	4	7				
6	2	2	6			
7	0	2	5	6	9	9
8	5	7	9			

Based on above data, which of the following statement is/are true?

(I) The range is 57
(II) The median is 71
(III) The mean is 66

a) Statement (I) only	**b) Statement (II) only**
c) Statement (III) only	d) All statements are correct

215) Which of the following is True about mean?

(i) Most repeated value
(ii) The best single figure to describe data

a) Statement (I) only	**b) Statement (II) only**
c) Statement (III) only	d) All statements are correct

216) Which of the following statements is/are true about population?
(i) It is desired to consider all the population in an observation.
(ii) In an observation we consider all of the population.

a) Only statement 1 is correct	b) Only statement 2 is correct
c) Both statements are correct	**d) None of the above are correct**

217) Which of the following statements is/are true about discrete data?
(i) There is no gap between them.
(ii) Mid-point is calculated by dividing the upper and lower limit in each class

a) Only statement 1 is correct	b) Only statement 2 is correct
c) Both statements are correct	**d) None of the above are correct**

218) Which of the following statements is/are true about discrete data?
(i) There is gap between them.
(ii) Mid-point is calculated by dividing the upper and lower limit in each class

a) Only statement 1 is correct	b) Only statement 2 is correct
c) Both statements are correct	d) None of the above are correct

219) Which of the following statements is/are correct?
(i) A grouped frequency distribution of discrete data has gaps between the classes.
(ii) In class boundaries, there is no gap between the classes.

a) Only statement 1 is correct	b) Only statement 2 is correct
c) Both statements are correct	d) None of the above are correct

220) Which of the following is correct?

a) The data which is collected specifically for ongoing investigation is called raw data	b) Results of sampling enquiries or a census is called primary data
c) The data which is relevant to the investigation but was collected previously for some other purpose is called secondary data	d) Both (b) and (c)

221) If the peak of the histogram is in the middle and the frequencies on either side are similar to each other, the distribution is said to be:

a) Normal	b) Balanced
c) Binomial	**d) Symmetrical**

222) Which of the following statements is correct?

a)	An Ogive is the graph of a cumulative frequency distribution
b)	Median of a grouped frequency distribution can be found by constructing an Ogive
c)	An Ogive is constructed by joining the mid points of the top of each rectangle histogram with straight lines
d)	**Both (a) and (b)**

223) Which of the following statements is correct?

a) A frequency polygon can be constructed by joining two symmetrical ogives
b) A histogram can be converted to an ogive by joining the mid-point of the top of each of its rectangle with a straight line
c) An ogive is the graph of a cumulative frequency distribution
d) Both (a) and (b)

224) A batter scored following runs in eleven T20 matches played in a calendar year:
35, 15, 51, 28, 0, 3, 85, 45, 30, 0
The mode of his above scores is 0

225) A population may be defined as including all people or items with a characteristic that a researcher wises to understand.

a) True	b) False

226) Which two of the following statements are correct regarding construction of a frequency distribution?

a) It sometime leads to provide incorrect data	b) It is the most common method of summarizing data
c) It begins by recording the number of times a particular value occurs	**d) It is the basis for construction of a percentage distribution**

227) Which two of the following statements are correct?

a) An ogive is the graph of a cumulative frequency distribution	**b) Median of a grouped frequency distribution can be found by constructing an ogive.**
c) An ogive is constructed by joining the mid points of the top of each rectangle of a histogram with straight lines	d) An ogive is the least desirable method of presentation of data

228) Consider the following statements about mean.

i. It must be one of the values found in the data

ii. In case of ungrouped data, there may be more than one mean

Which of the above statement is/are correct?

a) Only statement 1 is correct	b) Only statement 2 is correct
c) Both statements are correct	**d) None of the above are correct**

229) Construction of frequency distribution:

a) Helps in deleting the data	**b) Begins by recording the number of times a particular value occurs**
c) Is the only way to assess mode od population.	d) All of the above

230) Which of the following is correct?

a) Results of sampling enquiries or a census is called raw data	b) The data which is collected specifically for the ongoing investigation is called primary data
c) The data which is stored after classification is called secondary data	**d) Both (a) and (b)**

231) Which of the following is correct?

a) Results of sampling enquiries or a census is called raw data	b) The data which is collected specifically for the ongoing investigation is called primary data
c) The data which is stored after classification and summation is called secondary data	**d) Both (a) and (b)**

8 Statistical measures of data

232) Calculate variance the following data?

X	0	1	2	3	4
F	2	3	5	7	9

a) 1.26	**b) 1.60**	c) 2	d) 4

233) Geometric Mean of 7, 14 & 21?

a) 11.45	**b) 12.72**	c)14	d) 45.33

234) ∑(x—x)^2 =0.555 & n=5, find standard deviation?

a) 0.333	b) 0.003
c) 0.111	d) Insufficient Information

235) Calculate the Harmonic Mean of the following data 10,12, & 15

a) 12.33	b) 12.16	c) 12.31	**d) 12**

236) Calculate the Harmonic Mean of the following data 11,13, & 15

a) 12.33	b) 12.16	c) 12.31	**d) 12.80**

237) The standard deviation of the data given below is?

x	0	1	2	3	4
F	8	7	3	1	1

a. 1.095	b. 1.2	c. 1.44	d. 1

238) The variance of the data given below is?

X	0	1	2	3	4
F	8	7	3	1	1

a) 1.095	**b) 1.2**	c) 2.098	d) 4.4

239) The scores obtained by six students in a set of examination are 80, 40, 50, 72, 45 and 81. These scores are changed by 15% What will be the effect of these changes on the mean and standard deviation?

a) Mean and standard deviation will remain unchanged
b) Mean will increase by.15%Wbut standard deviation will remain unchanged
c) Mean and standard deviation both will increase by 15%
d) Mean will remain unchanged but standard deviation will increase by 15%

240) Which of the following statements is/are correct?

1. Mean can never be smaller than 1st quartile.
2. Mode can never be smaller than 1st quartile.

a) Both statements are correct	**b) Both statements are not correct**
c) Only statement (1) is correct	d) Only statement (2) is correct

241) Which of the following statement is/are correct?

1. Variance can never be larger than standard deviation
2. Variance is independent of change of scale

a) Both statements are correct	**b) Both statements are not correct**
c) Only statement (1) is correct	d) Only statement (2) is correct

242) Find the inner median if first quartile is 128 less than median and upper quartile is 256 more than median?

a) 256	b) 384
c) 128	**d) Impossible to determine.**

243) The mean of 11 numbers is 7. One of the numbers i.e 17 is deleted. The mean of the remaining 10 numbers is________

244) Starting salaries of a group of fresh graduates is as follows:

45,500, 50,000 48,000 60,000 62,000 55,000 58,000 and 49,000

Median of above salaries is:

a) 50,000	b) 53,437.50
c) 55,000	**d) 52,500**

245) Team A scored an average of 205 runs in twenty one-day international matches with a standard deviation of 10 whereas Team B scored an average of 190 runs in the same number of matches with a standard deviation of 8.

It may be concluded that Team A is more consistent than Team B

a) True	b) False

246) Mean of 38 values is 62, mean of 10 values is 57, find the mean of remaining 28 values

Answer

Using combined mean formula

$$\overline{Xc} = \frac{n_1 \overline{X}_1 + n_2 \overline{X}_2}{n_1 + n_2}$$

$$62 = \frac{10 \times 57 + 28 \times \overline{X}_2}{38}$$

$$\overline{X}_2 = 63.786$$

247) A batsman scored following runs in ten T20 matches played in a calendar year. 35,15,51,28,0,3,35,20,45,30. The mode of his scores is:

a) 0	**b) 35**
c) 45	d) 3

248) A batsman scored following runs in ten T20 matches played in a calendar year. 35,15,51,28,0,3,85,20,15,30. The mode of his scores is:

a) 0	**b) 15**
c) 45	d) 3

249) Immaculate Enterprise Assesses performance of its staff on annual basis assigning weightages to their performance scores. Following are the scores of two of its managers.

	Personal Growth	**Diversified skills**	**Appraising job results**	**Salaries/bonus**
Weight	80	80	64	75
Manager 1	200	170	50	20
Manager 2	60	80	170	140

Find the weighted arithmetic mean of mangers respectively?

a) 114.72 & 108.96	b) 108.96 & 114.72	c) 896.72 & 554.6	d) 554.6 & 896.72

250) Immaculate Enterprise Assesses performance of its staff on annual basis assigning weightages to their performance scores. Following are the scores of two of its managers.

	weightage	**Manager 1**	**Manager 2**
Annual goals achieved	5	75	80
Client Satisfaction	5	75	80
Addition of new clients	3	40	35
Initiatives taken	2	20	15

The weightage arithmetic mean of Manager 1 and Manager 2 are:

a) 62.33 and 60.67	b) 52.50 and 51.5
c) 60.67 and 62.33	d) 52.50 each

251) In order to decide whether to use z-test or t-test, we need to consider:

a) Size of sample only	**b) Size of sample and population variance**
c) Either size of sample or population variance	d) Either size of sample or population standard deviation

252) Consider the following data set: 11,19,19,20,21,24,25,25,36

The lower quartile of the data is:

a) 20	b) 25
c) 19	d) 24

253) Consider the following data: 2,5,7,6,9

The coefficient of variance of the above data is:

a) 36.94%	**b) 39.92%**
c) 41.57%	d) 60.08%

254) Consider the following data:

X	0	1	2	3	4
F	8	7	3	1	1

The variance of the above data is:

a) 1.095	**b) 1.2**
c) 2.098	d) 4.4

255) A survey of 316 randomly selected patients in a hospital, produced the following data:

Hospital Stays	1-3	4-6	7-9	10-12	13-15	16-18	19-21	22-24	25-27
Frequency (patients)	06	12	110	103	42	25	13	04	01

The mean and Standard Deviation of the data is:

a) 11 and 4	b) 11 and 5
c) 11and 3	d) 12 and 2

256) A survey of 316 randomly selected patients in a hospital, produced the following data:

Hospital Stays	1-3	4-6	7-9	10-12	13-15	16-18	19-21	22-24	25-27
Frequency (patients)	125	115	91	59	70	25	07	02	01

The mean and Standard Deviation of the data is:

a) 24.96 and 7.72	**b) 7.72 and 5.00**
c) 7.72 and 24.96	d) 5.00 and 7.72

257) Age distribution of employees in Dynamic Corporation is as follows:

Age in Years	22-26	26-30	30-34	34-38	38-42	42-46	46-50
No. of Employees	6	10	8	5	7	6	3

Find the Standard Deviation:

a) 7.3708	b) 6.9397
c) 6.1397	d) 6.3708

258) Age distribution of employees in Dynamic Corporation is as follows:

Age in Years	22-26	26-30	30-34	34-38	38-42	42-46	46-50
No. of Employees	6	10	8	5	7	1	3

Find the Standard Deviation:

a) 7.1397	**b) 6.9397**
c) 6.7397	d) 6.5397

259) The following data shows the weight (in grams, round to the nearest gram) of 35 randomly picked oranges from a farm:

155	161	164	166	168	170	172	172	173	175
177	178	178	179	181	182	182	184	186	188
189	192	195	196	197	198	203	206	208	209
210	214	218	221	243					

The mean and median of the above data is:

a) 186.50 and 182	b) 184.50 and 183
c) 188.29 and 184	d) None of the above

260) Team A scored an average of 205 runs in twenty-one-day international matches with a standard deviation of 10 whereas Team B scored an average of 190 runs in same one-day international matches with a standard deviation of 8. Which of the following is correct?

a) Team A is more consistent	**b) Team B is more consistent**
c) Both teams are equally consistent	d) Consistency cannot be determined from the above information

261) Detail of leaves availed during 20X8 by top three students in a class is as follows:

Name of Student	Taha	Amir	Zahid
Number of Leaves	7	5	3

Harmonic mean of the number of leaves taken by the above students is:

a) 5.00	b) 0.68

c) 0.20	**d) 4.44**

262) The mean and standard deviation of a sample of 100 values were found to be 104 and 4.7 respectively. Later, errors were discovered in three records enumerated below:

Sr. no.	Correct values (as per original record)	Amount taken (for computation)
58	151	115
72	78	87
89	98	89

The correct mean and standard deviation is:

a) 105.21 and 6.2	b) 107.5 and 7.2
c) 104.36 and 6.7	d) None of the above

263) Which of the following statements is correct as regards the mode of a grouped frequency distribution?

a) Modal class can be identified by preparing histogram	b) Mode is the mid-point of the Modal class
c) Modal class is the Mode of grouped frequency distribution	d) Both (a) and (b)

264) Calculate mode of the following data:
15, 16, 20, 20, 20, 21, 21, 21, 21, 23, 23, 28, 28, 28, 28, 28, 29, 29, 29, 33

a) 21	b) 29
c) 21 and 28	**d) 28**

265) A sample survey conducted by an organisation obtained the following data on the average number of items that persons in the various age group visit a physician each year:

Age Group (years)	Number of persons In the sample	Mean number of visits
Less than 5	50	2.1
5—20	115	1.6
21—60	155	2.6
61 and over	90	3.5

Calculate the mean number of visits to the physician:

a) 2.456	b) 2.656
c) 2.896	d) None of the above

266) If the median is 49.21 and the two quartiles are 37.15 and 61.27, what can be said of the skewness?

a) Distribution is positively skewed	b) Distribution is negatively skewed
c) Distribution is symmetrical	d) None of the above

267) Jamal got a rise of 12%, 20% and 18% in 2011, 2012 and 2013 respectively. The average annual increase rate is:

a) 16.5%	**b) 16.62%**
c) 17.1%	d) 17.33%

268) Detail of minor claims of an automobile insurance company is as follows:

Claims (Rs.)	1-1,000	1,001-2,000	2,001-3,000	3,001-4,000	4,001-5,000
No. of Claims	5	30	60	70	80

The standard deviation and variance for the insurance companies above data is:

a) 3,276 and 57	b) 1,093 and 33
c) 1,194,502 and 1,093	**d) 1,093 and 1,194,502**

269) Which of the following statements as regards to variance is/are correct?
(i) It is always smaller than the standard deviation
(ii) It can never be negative.

a) Only statement 1 is correct	**b) Only statement 2 is correct**
c) Both statements are correct	d) None of the above are correct

270) Which of the following statements as regards to variance is/are correct?
(i) It can never be smaller than the standard deviation
(ii) It can never be zero.

a) Only statement 1 is correct	b) Only statement 2 is correct
c) Both statements are correct	d) None of the above are correct

271) If in a frequency distribution, mode<median<mean then the frequency distribution is

a) Symmetrical around the median	b) Symmetrical around the mean
c) Skewed to left (negatively skewed)	**d) Skewed to right (positively skewed)**

272) If a frequency distribution is skewed to the left (negatively skewed), then:

a) Mean<median>mode	b) Mean>median>mode
c) Mode>median> Mean	d) Arithmetic mean < geometric mean < harmonic mean

273) Following is the data related to number of persons per house in a village town:

No. of persons per house	1	2	3	4	5	6	7	8	9	10
No. of houses	25	114	120	90	50	41	20	12	3	2

The mean, median and modal number of persons per house are:

a) 3.67, 3, 3	b) 3.62, 4, 5
c) 3.67, 3, 5	d) 3.42, 3, 3

274) Which of the following statement is CORRECT about the median?

a) The upper quartile is also called the median
b) It is a measure of central tendency
c) It is middle value no matter the data is arranged in any order
d) The position of the median can be found by using the expression $\frac{3(n+1)}{2}$

275)

Class boundary	10—12	12—14	14—16	16—18	18—20
Frequency	14	26	42	38	8

The geometric mean of the above data is:

a) 1.17	b) 14.53
c) 14.70	**d) 14.87**

276) Starting salaries of a group of fresh graduates are as follows:
45,000; 47,500; 52,000; 52,500; 52,500; 56,000; 56,500; 57,000
Find median from above data:

a) 52,000	b) 52,375
c) 52,500	d) 57,000

277) Starting salaries of a group of fresh graduates are as follows:
45,000; 47,500; 52,000; 52,500; 52,500; 56,000; 56,500; 57,000; 59,000; 60,000
Find median from above data:

a) 52,000	b) 52,500
c) 54,250	d) 56,000

278) Starting salaries of a group of fresh graduates are as follows:
63,000; 70,000; 68,000; 72,000; 78,000; 63,000; 65,000; 75,000;
Find median and mode from above data:

it is not the exact ans but is the closest answer

a) 63,000 and 70,000	b) 70,000 and 63,000
c) 71,000 and 63,000	d) 72,000 and 63,000

279) Which of the following statements is/are correct?

1. Range is a measure of dispersion.
2. Semi-inter Quartile Range measure of dispersion

a) Only statement 1 is correct	b) Only statement 2 is correct
c) Both statements are correct	d) None of the above are correct

280) Which of the following statements is/are correct?

I. Semi-inter quartile Range is a measure of dispersion.

II. Percentile is a measure of Dispersion

a) Only statement 1 is correct	b) Only statement 2 is correct
c) Both statements are correct	d) None of the above are correct

281) Which of the following statements is/are correct?

I. Range is a measure of dispersion.

II. Percentile is a measure of Dispersion

a) Only statement 1 is correct	b) Only statement 2 is correct
c) Both statements are correct	d) None of the above are correct

282) Which of the following statement is correct?

1. Quartile is a measure of dispersion.
2. Percentile is a measure of dispersion.

a) Only statement 1 is correct	b) Only statement 2 is correct
c) Both statements are correct	**d) None of the above are correct**

283) Which of the statement as regard to variance is/are correct?

i) It can never be smaller than standard deviation

ii) It can never be zero

a) Both statements are not correct	b) Both statements are correct
c) Only statement 1 is correct	d) Only statement 2 is correct

9 *Indices*

284) $\Sigma P_0 Q_o = 70.5,\ \Sigma P_1 Q_1 = 72,\ \Sigma P_1 Q_o = 60,\ \Sigma P_0 Q_1 = 73$

Find the Fisher's Price Index?

a. 107.65%	**b. 91.61%**	c. 109.15%	d. 92.89%

285) Which of the following about fisher index is true?

a) Fisher index is always greater than Lapeyre's and Paasche index
b) Fisher index is always less than Lapeyre's and Paasche index
c) Fisher index is either greater than or less than Lapeyre's and Paasche index
d) Fisher index always lies between Lapeyre's and Paasche index

286) The prices of commodity in different years is given below:

Years	Prices
2010	49
2011	53
2012	58
2013	62

Determine the chain indices in the above case:

a) 100,105.5,118.40,112.80	b) 100,92.50,91.40,93.55
c) 100,108.16,109.43,106.90	d) 100,105.50,116.50,114.50

287) The price relatives of three commodities are given below:

Year	A	B	C
2013	100	105	102
2014	106	100	100

The chain indices for each commodity are:

a) 94, 105 and 102 respectively	**b) 106, 95 and 98 respectively**
c) 106, 105 and 102 respectively	d) 106, 95 and 102 respectively

288) Following CPI has been computed taking 2008 as base year.

Year	CPI
2008	104.98
2009	100.00
2010	116.19
2011	115.11
2012	132.01

Yearly inflation/ deflation for the above data shall be?

a) (4.98%), 5.83%, 7.75%, 12.80%	b) 4.98%, 6.19%, 8.41%, 14.69%
c) (4.74%), 6.19%, 8.40%, 14.68%	d) 4.74%, 6.19%, 8.40%, 14.68%

289) Consider following data

Ingredients	Jan 20x2 (Base Period)		Dec 20x5 (current period)	
	Price per kg	Kg per unit	Price per kg	Kg per unit
A	3.00	10.00	3.95	11.00
B	9.00	3.00	9.90	2.50
C	1.00	2.00	0.95	3.00
D	2.00	2.00	4.50	5.00

Using Paasche Price index as at Dec 20x5, which of the following statement is correct?

a) Prices have risen by 8.73% between Jan 20x2 and Dec 20x5
b) Prices have risen by 16.79% between Jan 20x2 and Dec 20x5
c) Prices have risen by 27.14% between Jan 20x2 and Dec 20x5
d) Prices have risen by 36.57% between Jan 20x2 and Dec 20x5

290) Chemical Master Company (CMC) produces a special industrial chemical that is a blend of four chemical ingredients. The prices at the beginning and the end of year of each material and quantities required to make one unit of finished product are given below:

Ingredients	**Jan 20X2 (Base Period)**		**Dec 20X5 (Current Period)**	
	Price per kg	**Kg per unit**	**Price per kg**	**Kg per unit**
A	2.50	10.00	3.95	11.00
B	8.75	3.00	9.90	2.50
C	0.99	2.00	0.95	3.00
D	4.00	2.00	4.50	5.00

Using Lapeyre price index as at Dec 20X5, which of the following statement is correct?

a) Prices have risen by 30.82% between Jan 20X2 and Dec 20X5	b) Prices have risen by 23.56% between Jan 20X2 and Dec 20X5
c) Prices have risen by 22.67% between Jan 20X2 and Dec 20X5	d) Prices have risen by 29.31% between Jan 20X2 and Dec 20X5

291) Chemical Master Company (CMC) produces a special industrial chemical that is a blend of four chemical ingredients. The prices at the beginning and the end of year of each material and quantities required to make one unit of finished product are given below:

Ingredients	**Jan 20X2 (Base Period)**		**Dec 20X5 (Current Period)**	
	Price per kg	**Kg per unit**	**Price per kg**	**Kg per unit**
A	3.00	10.00	3.95	11.00
B	9.00	3.00	9.90	2.50
C	1.00	2.00	0.95	3.00
D	2.00	2.00	4.50	5.00

Using Laspeyre price index as at Dec 20X5, which of the following statement is correct?

a) Prices have risen by 30.82% between Jan 20X2 and Dec 20X5	b) Prices have risen by 23.56% between Jan 20X2 and Dec 20X5
c) Prices have risen by 22.67% between Jan 20X2 and Dec 20X5	d) Prices have risen by 29.31% between Jan 20X2 and Dec 20X5

292) Which of the following statements is correct about Laspeyre price index?

a) It fails to account for the fact that people will buy less of those items which have risen in price
b) The denominator in the Laspeyre price index has to be recalculated every year to take account of the most recent quantities consumed.
c) It is based on most recent quantities purchased.
d) It tends to understate inflation

293) Which of the following is correct about Laspeyre price index?

a) It has a focus which is biased to the cheaper items bought by consumers as a result of inflation.
b) The denominator in the Laspeyre price index has to be recalculated every year to take account of the most recent quantities consumed.
c) It is based on quantities bought in the base year
d) It tends to understate inflation

294) Which of the following statements about Laspeyre's Price Index is/are correct?

(i) It tends to overstate inflation.

(ii) The denominator does change from year to year.

a) Only statement 1 is correct	b) Only statement 2 is correct
c) Both statements are correct	d) None of the above are correct

295) Which of the following statements about Laspeyre's Price Index is/are correct?

(i) It tends to overstate inflation.

(ii) The denominator does not change from year to year.

a) Only statement 1 is correct	b) Only statement 2 is correct
c) Both statements are correct	d) None of the above are correct

296) Which of the following is correct about Laspeyre price index?

a) It has a focus which is biased to the cheaper items bought by consumers as a result of inflation.
b) The denominator in the Laspeyre price index has to be recalculated every year to take account of the most recent quantities consumed.
c) It is based on the most recent quantities purchased
d) It tends to overstate inflation

297) Which of the following is correct about Laspeyre price index?

a)	It has a focus which is biased to the cheaper items bought by consumers as a result of inflation.
b)	**The denominator in the Laspeyre price index does not change from year to year.**
c)	It is based on the most recent quantities purchased
d)	It tends to understate inflation

298) Which of the following is correct about Paasche index?

a)	It fails to account for the fact that people will buy less of those items which have risen in prices.
b)	The denominator in the Paasche price index does not change from year to year.
c)	**It is based on the most recent quantities purchased**
d)	It tends to overstate inflation

299) Compute Laspeyres Price Index for the following data using 2002 as base:

Ingredients	Price in 2002	Price in 2007	Quantity in 2002
A	140	220	40
B	120	180	25
C	80	110	60

a) 147.21	**b) 148.51**
c) 149.50	d) 146.50

300) Which of the following statement is/are correct?

(i) In Laspeyre price index, the denominator does not change from year to year.

(ii) In Laspeyre price index, the denominator has to be recalculated every year to take account of the most recent quantities consumed

a) Only statement 1 is correct	b) Only statement 2 is correct
c) Both statements are correct	d) None of the above are correct

301) Which of the following statement is/are correct?

(i) In Laspeyre price index, the denominator does change from year to year.

(ii) In Paasche's price index, the denominator has to be recalculated every year to take account of the most recent quantities consumed

a) Only statement 1 is correct	**b) Only statement 2 is correct**
c) Both statements are correct	d) None of the above are correct

302) Which of the following statement is/are correct?

(i) In Laspeyre price index, the denominator does not change from year to year.

(ii) In Paasche's price index, the denominator has to be recalculated every year to take account of the most recent quantities consumed

a) Only statement 1 is correct	b) Only statement 2 is correct
c) Both statements are correct	d) None of the above are correct

303) Which of the following statement is/are correct?

(i) In Laspeyre price index, the denominator does not change from year to year.

(ii) Laspeyre price index is used more than the Paasche price index in practice

a) Only statement 1 is correct	b) Only statement 2 is correct
c) Both statements are correct	d) None of the above are correct

304) The index that uses quantities of base period as weights, is known as:

a) Laspeyre's index	b) Paasche's index
c) Fisher's index	d) Both (a) and (b)

305) Paasche price index fails to account for the fact that people will buy less of those items which have risen in price.

c) True	**d) False**

306) The prices of commodity in different years are given below:

Year	Prices
2010	49
2011	53
2012	58
2013	62

The chain indices in the above case would be:

a) 100,92.50,91.40,93.55	**b) 100, 108.16, 109.43, 106.90**
c) 100, 105.50, 118.40, 112.80	d) 100, 105.50, 116.50, 114.50

307) The price of a juicer machine during past five years is as follows:

Year	Price (Rs.)
20X7	3,700
20X8	4,800
20X9	4,500
20Y0	4,500
20Y1	5,300

Keeping 20X9 as base year, simple index numbers relative to price for the given period would be:

a) 100, 121.62, 129.73, 135.14, 143.24
b) 121.62, 100, 93.75, 90, 84.91
c) 100, 82.22, 77.08, 74, 69.81
d) 82.22, 106.67, 100, 100, 117.78

308) The price of a juicer machine during past five years is as follows:

Year	Price (Rs.)
20X7	3,700
20X8	4,500
20X9	4,800
20Y0	5,000
20Y1	5,300

Keeping 20X8 as base year, simple index numbers relative to price for the given periodwould be:

a) 100, 121.62, 129.73, 135.14, 143.24
b) 121.62, 100, 93.75, 90, 84.91
c) 100, 82.22, 77.08, 74, 69.81
d) 82.22, 100, 106.67, 111.11, 117.78

309) The price of a Juicer Machine during past five years is as follows:

Year	Price (Rs.)
20X7	3,700
20X8	4,500
20X9	4,800
20Y0	5,000
20Y1	5,300

Keeping 20X7 as base year, simple index numbers relative to price for the given period would be:

a) 100, 121.62, 129.73, 135.14, 143.24	b) 121.62, 100, 93.75, 90, 84.91
c) 100, 82.22, 77.08, 74, 69.81	d) 82.22, 100, 106.67, 111.11, 117.78

310) Consumer Price index of a country in 2018 is 148 it increases in 2019 by 10%, in 2020 it decreases by 10%, then again increases by 10% in 2021 then again decreases by 10% in 2022. Find index of 2022.

a) 148	b) 149
c) 145	d) 154

10 Correlation and regression

311) If most of the points are clustered around a single point in a scatter diagram, it will most probably depict:

a) Linear relationship.	b) Positive relationship
c) Negative relationship	**d) No relationship.**

312) Which of the following statement is /are correct about correlation coefficient?
1. It is always from 0 to +1
2. It is always from 0 to -1

a) Only statement 1 is correct	b) Only statement 2 is correct
c) Both statements are correct	**d) None of the above are correct**

313) Which of the following is true about regression lines?
1. It is used to obtain line of best fit between two variables.
2. We can calculate fixed cost from it but not variable cost.

a) Only statement 1 is correct	b) Only statement 2 is correct
c) Both statements are correct	d) None of the above are correct

314) When correlation coefficient is closer to one then:

a) No correlation	b) High correlation
c) Moderate correlation	**d) Linear correlation**

315) When on scatter diagram all points cluster around a single point is called:

a) No correlation	b) High correlation
c) Moderate correlation	d) Linear correlation

316) Desi Restaurant collected the following data about advertising and sales for five months:

Advertising Expense (Rs. 000)	10	40	20	35	50
Sales (Rs. 000)	125	535	180	415	560

Equation of a line of best fit for the above data will be:

a) Y= - 17.06+12.26x	b) Y=17.06-12.26x
c) Y=-17.06x+12.26	d) Y=17.06x-12.26

317) Which of the following is correct?

a) If there is a strong relationship between two variables, the points on the scatter diagram concentrated around a curve
b) Linear regression analysis is used to calculate values of "a" and "b" in the linear cost equation
c) The standard regression equation is y= a – bx
d) Both (a) and (b)

318) Which of the following is correct?

a) If there is a strong relationship between two variables, the points on the scatter diagram would be concentrated around a curve
b) Linear regression analysis can be used to estimate fixed costs and variable cost per unit from historical total cost and production data.
c) The standard regression equation is y= a – bx
d) Both (a) and (b)

319) Which of the following is correct?

a) If there is a strong relationship between two variables, the points on the scatter diagram would be concentrated around a curve
b) The standard regression equation is y= a – bx
c) Scatter diagram leads to correct conclusions even if there are few data points
d) Both (a) and (b)

320) Which of the following statements about scatter diagram is/are correct?
(i) It is important to establish which variable is independent before plotting
(ii) It might indicate a relationship where there is none

a) Only statement 1 is correct	b) Only statement 2 is correct
c) Both statements are correct	d) None of the above are correct

321) Which of the following statements about scatter diagram is/are correct?

(i) It is least important to establish which variable is independent

(ii) It might indicate a relationship where there is none.

a) Only statement 1 is correct	**b) Only statement 2 is correct**
c) Both statements are correct	d) None of the above are correct

322) Which of the following statements about scatter diagram is/are correct?

(i) It leads to correct conclusion even if there are only few data points.

(ii) It might indicate a relationship where there is none.

a) Only statement 1 is correct	**b) Only statement 2 is correct**
c) Both statements are correct	d) None of the above are correct

323) Which of the following statements is correct as regards scatter diagrams?

a) It is least important to establish which variable is independent before plotting the scatter diagram	b) It leads to correct conclusions even if there are only few data points.
c) It can show relationship between more than two variables	**d) It may indicate a relationship where there is none**

324) Which of the following statements is correct as regards scatter diagrams?

a) It can show relationship between more than two variables	b) It leads to correct conclusions even if there are only few data points.
c) It is important to establish which variable is independent before plotting the scatter diagram	d) Both (b) and (c)

325) Which of the following statement is/are correct where Regression is used?

(i) It is very important in accounting and other business operations

(ii) It is widely used in economics & business

a) Only statement 1 is correct	b) Only statement 2 is correct
c) Both statements are correct	d) None of the above are correct

326) Which of the following statement is/are correct where Regression is used?

(i) It tells fixed cost but not variable cost per unit.

(ii) It shows the extent of relationship between two variables

a) Only statement 1 is correct	b) Only statement 2 is correct
c) Both statements are correct	**d) None of the above are correct**

327) A medical research company has developed the following equation for regression line of y on x (line of best fit) for a particular age group y= 6.93 +0.38x where x represent height in centimeters and y represents weight in kilogram. Using above equation, we can say that

a) For each centimetre increase in height, weight will decrease by 0.38 kilogram
b) For each centimetre increase in height, weight will increase by 0.38 kilogram
c) For each centimetre increase in height, weight will increase by 6.55 kilogram
d) For each centimetre increase in height, weight will decrease by 6.55 kilogram

328) Consider following data

Towns	Bee	Cee	Dee	Gee	Jay	Kay	Pee	Tee
Police strength	140	130	220	150	140	150	180	160
No. of crimes per Month	95	110	80	75	90	120	100	110

The equation for regression line of y on x (line of best fit) for the above data is:

$$Y=129.25-0.2x$$

Using the above regression equation, which of the following statement is correct?

a) Police of Bee town is more efficient than police of Cee town
b) Police of Dee town is more efficient than police of Gee town
c) Police of Kay town is more efficient than police of Pee town
d) Both (a) and (b)

329) The Citizen Police Liaison Committee of Port City has gathered following information from the various towns of the city:

Towns	**East**	**West**	**South**	**North**	**Central**	**Costal**	**Upper**	**Lower**
Police strength	120	150	130	125	156	150	130	160
No. of crimes per Month	105	90	108	108	155	180	130	125

The equation for regression line of y on x (line of best fit) for the above data is:

a) Y=31.825+1.123x	b) Y=3.840+0.866x
c) Y=110.533+0.237x	d) Y=3.840+0.237x

330) The Citizen Police Liaison Committee of Utopia City has gathered following information from the various towns of the city:

Towns	Bee	Cee	Dee	Gee	Jay	Kay	Pee	Tee
Police strength	120	150	230	235	156	150	130	160
No. of crimes per Month	95	110	90	55	90	180	130	110

The equation for regression line of y on x (line of best fit) for the above data is:

Y=185.031 – 0.466x

Using the above regression equation, which of the following statement is correct?

a) Police of Cee town is more efficient than police of Bee town
b) Police of Dee town is more efficient than police of Gee town
c) Police of Kay town is more efficient than police of Pee town
d) Police of Jay town is more efficient than police of Tee town

331) $If\ \ \Sigma X = 2000,\ \Sigma Y = 1020,\ \Sigma X^2 = 10{,}000,\ \ \Sigma Y^2 = 750,\ \Sigma XY = 15{,}000\ \ and\ \ n = 100$

The line of regression of Y on X is:

a) 5.6+0.17x	**b) 6.6+0.18x**
c) 6.6+0.15x	d) 9.6+0.18x

332) $If\ \ \Sigma X = 1{,}239,\ \ \Sigma Y = 79,\ \Sigma X^2 = 568{,}925,\ \ \Sigma Y^2 = 293,\ \ \Sigma XY = 17{,}233\ \ and\ \ n = 100$

Find line y on x and x on y and their point of intersection

a)	**b)**
c)	d)

333) $If\ \ \Sigma X = 75,\ \Sigma Y = 225,\ \ \Sigma XY = 2{,}851,\ \ \Sigma X^2 = 625,\ \ and\ \ n = 6$, then equation for regression line of y on x (line of best fit) would be:

a) Y= –39 +0.12x	b) Y=39 – 0.12x
c) Y=39+0.12x	d) Y= –39 – 0.12x

334) $If\ \ \Sigma X = 173,\ \ \Sigma Y = 613,\ \Sigma XY = 11{,}965,\ \Sigma X^2 = 4{,}119,\ \ and\ \ n = 10$, then equation for regression line of y on x (line of best fit) would be:

a) Y=40.402+20.898x	b) Y=-61.30+20.898x
c) Y=40.402+1.208x	d) Could not be calculated as value of ΣY^2 is not given

335) $If\ \ \Sigma X = 65,\ \ \Sigma Y = 315,\ \Sigma XY = 1985,\ \Sigma X^2 = 685,\ \ and\ \ n = 5$. Then equation for regression line y on x?

a) Y=108.47+13.19x	b) Y= – 108.47 – 13.19x
c) Y= – 108.47+13.19x	d) Y=108.47–13.19x

336) $If\ \ \Sigma X = 173,\ \ \Sigma Y = 613,\ \Sigma XY = 11{,}965,\ \Sigma X^2 = 4{,}119,\ \ and\ \ n = 10$, then equation for regression line of y on x (line of best fit) would be:

a) Y=40.40+20.90x	b) Y=-61.30+20.90x
c) Y=40.40+1.21x	d) Could not be calculated as value of ΣY^2 is not given

337) $If\ \ \Sigma X = 58,\ \ \Sigma Y = 313,\ \Sigma XY = 3{,}015,\ \Sigma X^2 = 594,\ \ and\ \ n = 6$ then the equation for regression line of y on x (line of best fit) would be:

a) X=-0.32+55.26y	b) Y=55.26+0.32x
c) Y=0.32-55.26x	d) Y=55.26-0.32x

338) Which of the following statements as regards to the value of the correlation coefficient is/are correct?

(i) The coefficient of determination is the square of the correlation coefficient
(ii) The value of the coefficient of determination must always be in the range 0 to +1.

---value of co-relation coefficient ranges from 0 to +,- 1
---value of corfficient of determination ranges from 0 to 1 because it is the square of coefficient of corelation and square is not negative

a) Only statement 1 is correct	b) Only statement 2 is correct
c) Both statements are correct	d) None of the above are correct

339) Find the coefficient of correlation between x and y if:
Regression line of x on y is: 5x-4y+2=0 and
Regression line of y on x is: x-5y+3=0

a) 0.4	b) -0.4
c) ±0.4	d) ±0.16

340) If a = -12.57 and b = 0.35 then equation for regression line of y on x (line of best fit) would be:

a) Y=-0.35+12.57x	b) Y=0.35-12.57x
c) Y=12.57- 0.35x	**d) Y= -12.57+0.35x**

341) If the equation for regression line of y on x (line of best fit) is y=16 - 1.5x, then by increasing every 1-unit of x the value of y would:

a) Decrease by 16 units	b) Decrease by 14.5 units
c) Decrease by 1.5 units	d) Increase by 14.5 units

342) If the equation for regression line of y on x (line of best fit) is y=32+0.5x, then for every unit increase in x, y would:

a) Increase by 32 units	b) increase by 16 units
c) increase by 48 units	**d) Increase by 0.5 units**

343) What will be the coefficient of correlation for a sample of 20 pairs of observations, given that:

$$\overline{X} = 2,\ \overline{Y} = 8,\ \Sigma X^2 = 180,\ \ \Sigma Y^2 = 1424\ \ and\ \ \Sigma XY = 404$$

a) 0.90	**b) 0.70**
c) 0.80	d) None of these

344) If the value of coefficient of determination i.e r^2=0.64, it means that:

a) 80% of variations in the value of y are explained by variations in the value of x	**b) 64% of variations in the value of y are explained by variations in the value of x**
c) 36% of variations in the value of y are explained by variations in the value of x	d) 20% of variations in the value of y are explained by variations in the value of x

345) If the coefficient of determination is a positive value, then the coefficient of correlation:

a) Must be positive	b) Must be negative
c) Must be zero	**d) Can be positive as well as negative**

346) Which of the following statements about scatter diagram is/are correct?
(i) It is least important to establish which variable is independent
(ii) It leads to correct conclusion even if there are only few data points.

a) Only statement 1 is correct	b) Only statement 2 is correct
c) Both statements are correct	**d) None of the above are correct**

347) Which of the following statements about scatter diagram is/are correct?
(i) Shows visual impression (relation) between variables.
(ii) It might indicate a relationship where there is none.

a) Only statement 1 is correct	b) Only statement 2 is correct
c) Both statements are correct	d) None of the above are correct

348) There is a perfect positive correlation when

a) All the data points lie on a scatter diagram in a discrete form	**b) all the data points lie in an exact straight line and a linear relationship exists between the two variables**
c) All the data points lie in an exact straight line but relationship, may or may not be linear	d) None of the above

349) Perfect positive correlation is denoted by value:

a) 1	b) 0
c) Between 0.90 and 1.0	d) 1 or -1

350) Value of r=0, indicates:

a) Perfect positive correlation	b) Perfect negative correlation
c) No correlation	d) None of the above

351) If 25% of variations in the value of y are explained by variations in the value of x then coefficient of correlation is:

a) 0.25	b) 0.75

c) 0.50 or -0.50	d) Either 0.25 or – 0.25

352) If 49% of variations in the value of y are explained by variations in the value of x then coefficient of correlation is:

a) 0.51	**b) 0.70**
c) 0.50 or -0.50	d) Either 0.49 or – 0.49

353) If the value of r2 is 0.25, it means that: (01 mark)

a) 75% of the variation in the value of y is explained by variation in the value of x
b) 50% of the variation in the value of y is explained by variation in the value of x
c) 25% of the variation in the value of y is explained by variation in the value of x
d) 0.25% of the variation in the value of y is explained by variation in the value of x

354) If covariance(x,y)=62, standard deviation (x) = 16 and standard deviation (y)=7, "r" will be equal to:

a) 0.6435	**b) 0.5536**
c) 0.6235	d) 0.5235

355) If all the points on scatter diagram lies on the regression line, then coefficient of correlation is?

a) 0	b) 1
c) -1	**d) 1 or -1**

356) If regression coefficient of y on x is 1.44 and the coefficient of correlation between x and y is 0.6, the regression coefficient of x on y will be

a) 0.84	b) 2.4
c) 0.25	d) 4

357) Find out the regression coefficient of y on x if regression coefficient of x on y is 0.25 and coefficient of correlation between x and y is 0.6:

a) 1.24	b) 1.44
c) 1.96	d) 2.02

358) The average runs scored by seven leading test cricketers during last calendar year are given below:

Average runs scored in 1st innings (x)	46	73	68	79	49	43	81
Average runs scored in 2nd innings (y)	31	55	65	62	85	36	53

The spearman's rank correlation coefficient for the runs scored in first and second innings is:

a) 0.7143	b) 0.1190
c) 0.2857	d) 0.8810

359) Find rank correlation of the following data

1	3	3	5	4	4
1	2	3	4	5	6

a) 0.7824	b)
c)	d)

360) $\Sigma(x-\bar{x})(y-\bar{y}) = 954$ and standard deviation of x is 21.5 and standard deviation of y is 10.61 r=0.524 then find n=?

a) 20	b) 7
c) 9	**d) 8**

361) Which of the following is correct?

a) The coefficient of determination is the square root of the correlation coefficient.	b) The coefficient of coefficient is the square of the correlation of determination.
c) The coefficient of correlation explains the variations in the value of y by variations in the value of x.	**d) The coefficient of determination explains how much variability of one factor can be caused by relationship to another related factor**

362) Which of the following statements regarding regression is correct?

a) If there is a weak relationship between two variables, the points on the scatter diagram would be concentrated around a curve	b) In linear regression, both variables are dependent to each other
c) The standard regression equation is Y=a-bx	**d) In linear regression, one variable is considered to be an explanatory variable, and the other is considered to be a dependent variable**

363) Which of the following statements regarding regression is correct?

a) If there is a weak relationship between two variables, the points on the scatter diagram would be concentrated around a curve	**b) If the curve of regression is a straight line, it is described as a linear regression line**.
c) The standard regression equation is Y=a-bx	d) In linear regression, both variables are dependent to each other

364) The correlation between height and weight for adults is +0.90. it depicts height on account of variation in weight is:

a) 90%	b) 45%
c) 10%	**d) 81%**

365) Following equations are given

X+2Y+5=0

X+3Y−10=0

Find coefficient of correlation

a) 0.8165	**b) – 0.8165**
c) 1.2247	d) -1.2247

366) From following information given, find coefficient of correlation.

X	3	5	8	11	9
Y	1	0	4	0	1

a) r= 0.038	b) r= 0.082
c) r=0.018	d) r= none of these

367) Which of the following is correct about coefficient of determination?

a) It ranges from 0 to -1	b) It is Square root of r
c) It ranges from 0 to +1	d) All of these

368) Find the coefficient of correlation between x and y if:

Regression line of x on y is: 5x-4y+2=0 and

Regression line of y on x is: x-5y+3=0

a) 0.4	b) -0.4
c) ±0.4	d) ±0.16

369) If r falls between 0.9 and 1.0 then which of the following is correct

a) There is strong correlation	b) There is no correlation
c) There is perfect correlation	d) None of these

370) $\overline{X} = 6,\ \overline{Y} = 10,\ \ \Sigma XY = 293,\ \ \Sigma X^2 = 266,\ \ n = 6,\ \ \Sigma Y^2 = 706$

Find coefficient of correlation

a) 0.9203	**b) –0.9203**
c) –0.2903	d) 0.2903

371) $\overline{X} = 6,\ \overline{Y} = 10,\ \ \Sigma XY = 293,\ \ \Sigma X^2 = 266,\ \ n = 6,\ \ \Sigma Y^2 = 706$

Find coefficient of correlation

a) 0.9203	**b) –0.9203**
c) –0.2903	d) 0.2903

372) Y = 1.96x + 15 (y on x), X = 0.45y + 7.16 (X on Y). Find Co-efficient of determination.

a) 0.94	b) 0.9983
c) **0.88**	d) 0.5569

373) If the Co-efficient of determination is equal to 1, then correlation Co-efficient is:

a) Must be equal to 1	b) Any value between – 1 and + 1
c) **Either – 1 or + 1**	d) Must be – 1

374) In regression, the sum of the residuals is always:

a) **0**	b) >0
c) <0	d) All of these

375) For 9 observations on supply (x) and price (y), following data was obtained $\Sigma(Y-127)=12$, $\Sigma(Y-127)^2=1006$ $\Sigma(X-90)=-25$, $\Sigma(X-90)^2=301$ $\Sigma(X-90)(Y-127)=469$ The estimated value of supply when the price is Rs 125/- comes to be:

a) **87.79**	b) 78.79
c) 79.87	d) 79.78

376) if the coefficient of correlation between x and y is -0.75 the SD of y is 5 and $\Sigma(x-\bar{x})(y-\bar{y})=-15$. The value of SD of x would be:

a) **4**	b) 16
c) 5	d) 6

377) For r2 = 0.6 the explained variation in dependent variable due to independent variable is:

a) **0.6**	b) 0.4
c) 0.36	d) 0.16

378) If the regression line is a perfect estimator of the dependent variable, then which of the following is false.

a) Co – efficient of determination is one	b) **Co – efficient of correlation is zero**
c) All the data points fall on regression line	d) None of these

379) There is a perfect positive correlation when

a) All the data points lie on a scatter diagram in a discrete form	b) **all the data points lie in an exact straight line and a linear relationship exists between the two variables**
c) All the data points lie in an exact straight line but relationship ,may or may not be linear	d) None of the above

380) Which of the following statements is correct as regards scatter diagrams?

a) It is least important to establish which variable is independent before plotting the scatter diagram	b) It leads to correct conclusions even if there are only few data points.
c) It can show relationship between more than two variables	**d) It may indicate a relationship where there is none**

381) Which of the following statements is correct as regards scatter diagrams?

a) It can show relationship between more than two variables	b) It leads to correct conclusions even if there are only few data points.
c) It is important to establish which variable is independent before plotting the scatter diagram	d) Both (b) and (c)

382) Which of the following is correct?

a) If there is a strong relationship between two variables, the points on the scatter diagram concentrated around a curve
b) Linear regression analysis is used to calculate values of "a" and "b" in the linear cost equation
c) The standard regression equation is y= a – bx
d) Both (a) and (b)

383) Which of the following is correct? (01)

a) If there is a strong relationship between two variables, the points on the scatter diagram would be concentrated around a curve
b) Linear regression analysis can be used to estimate fixed costs and variable cost per unit from historical total cost and production data.
c) The standard regression equation is y= a – bx
d) Both (a) and (b)

384) Which of the following statements is/are correct?

(i) There is strong relationship between two variables, the points on the scatter diagram would beconcentrated around a curve.

(ii) Linear regression analysis can be used to estimate fixed cost and variable cost per unit from historical totalcost and production data.

a) Only statement 1 is correct	b) Only statement 2 is correct
c) Both statements are correct	d) None of the above are correct

385) Which of the following is correct? (01)

a) If there is a strong relationship between two variables, the points on the scatter diagram would be concentrated around a curve
b) The standard regression equation is y= a – bx

c)	Scatter diagram leads to correct conclusions even if there are few data points
d)	Both (a) and (b)

386) Which two of the following are correct?

a) The value of coefficient of determination shows how much variation in the value of y is explained by variation in the value of x	b) If coefficient of correlation i.e r=0, it means there is a perfect correlation between x and y
c) The value of coefficient of determination is in range -1 to +1	**d) The perfect negative or positive coefficient of correlation cannot be achieved in real life scenario.**

387) Which of the following statement as regard to the value of the correlation coefficient is/are correct?

i. It always greater than zero

ii. It always lies in the range of -1 to +1

a) Only statement 1 is correct	**b) Only statement 2 is correct**
c) Both statements are correct	d) None of the above are correct

388) Which of the following statements about scatter diagram is/are correct?

(i) It leads to more accurate results if the collected data is atypical

(ii) we can draw it for two or more variables

a) Only statement 1 is correct	b) Only statement 2 is correct
c) Both statements are correct	**d) None of the above are correct.**

389) Which of the following statement regarding scatter diagram is/are correct?

I. It leads to more accurate results if the collected data is atypical

II. It might indicate a relationship where there is none

a) Only statement 1 is correct	**b) Only statement 2 is correct**
c) Both statements are correct	d) None of the above are correct

390) Which of the following statements about scatter diagram is/are correct?

1. Shows visual relation between variables
2. It might indicate a relationship where there is none

a) Only statement 1 is correct	b) Only statement 2 is correct
c) Both statements are correct	d) None of the above are correct

11 Probability concepts

391) A wallet contains 65 notes of Rs 1,000 and 35 notes of Rs. 500 Four notes are selected at random with replacement. Find the probability that sum of notes are Rs. 3,000?

a) 0.3156	b) 0.6844	**c) 0.3105**	d) 0.6895

392) A coin is rolled three times, find the probability that exactly one head will appear.

a) **0.375**	b) 0.625	c) 0.357	d) 0.875

393) If there are 5 Civic cars & 8 city cars. If two cars are selected, what is the probability that both are Civics?

a) **0.1282**	b) 0.8718
c) 5/13	d) 0.3571

394) There are 30 students in PRC class consisting of 18 boys and 12 girls. 2 boys and 2 girls are selected at random from this class, you are required to find how many different combinations are formed?

a) 27405	b) **10,098**
c) 40392	d) 657720

395) The value of 0! is equal to __1__ and 1! is equal to __1__.

396) In how many different ways can the letters of the word 'LEADING' be arranged in such a way that the vowels always come together?

a) 120	b) 360
c) 720	d) 5040

Answer

The word 'LEADING' has 7 different letters.

When the vowels EAI are always together, they can be supposed to form one letter.

Then, we have to arrange the letters LNDG (EAI).

Now, 5 (4 + 1 = 5) letters can be arranged in 5! = 120 ways.

The vowels (EAI) can be arranged among themselves in 3! = 6 ways.

Required number of ways = (120 x 6) = 720.

397) The events A and B are mutually exclusive if P(A)=0.5 and P(B)=0.4 then P (A or B) is:

a) 0.1	b) 0.54
c) 0.2	**d) 0.9**

398) A 4-digit pin code can begin with any number except 0 ,1,2. If repetition of the same digit is allowed the probability that the pin will begin with 3 is?

a) 1/7	b) 1/9
c) 1/8	d) 1/49

399) A 4-digit pin code can begin with any number except 0,1 & 2. If repetition of the same digit is allowed, the probability of a pin code begin with 5 and end with a 3 is?

a) 1/70	b) 1/90
c) 1/80	d) 1/49

400) If a consignment of 25 auto batteries 3 are defective. If a random sample of 5 batteries is selected the probability of having exactly 2 defective batteries is:

a) 0.053	b) 0.035
c) 0.12	**d) 0.087**

401) The table below describes the smoking habits of a group of asthma sufferers:

Gender	Non smokers	Light smokers	Heavy Smokers	Total
Men	353	42	49	444
Women	352	32	40	424
Total	705	74	89	868

If a person is randomly selected from the group, the probability that selected person is either Women or Light Smoker Male, is:

a) 0.5737	**b) 0.5369**
c) 0.5115	d) 0.0373

402) How many 3-digit numbers can be formed from the digits 1, 3, 4, 5, 7 and 9, which are divisible by 2 and none of the digits is repeated?

a) 6	**b) 20**

c) 30	d) 120

Answer

Since each desired number is divisible by 2, so we must have 4 at the unit place. So, there is 1 way of doing it.
The tens place can now be filled by any of the remaining 5 digits (1, 3, 5, 7, 9). So, there are 5 ways of filling the tens place.
The hundreds place can now be filled by any of the remaining 4 digits. So, there are 4 ways of filling it.
Required number of numbers = (1 x 5 x 4) = 20.

403) A 5digit Pin Code can begin with any number except 0 and 1. If repetition of the same digit is allowed then probability of a Pin Code beginning with a 7 and ending with an 8 is

a) 1/10	b) 1/100
c) 1/90	**d) 1/80**

404) How many 3-digit numbers can be formed from the digits 2, 3, 5, 6, 7,8 and 9, which are divisible by 5 and none of the digits is repeated?

a) 12	**b) 30**
c) 120	d) 720

405) How many 3-digit numbers can be formed from the digits 2, 3, 5, 6, 7 and 9, which are divisible by 5 and none of the digits is repeated?

a) 6	**b) 20**
c) 120	d) 360

Answer

Since each desired number is divisible by 5, so we must have 5 at the unit place. So, there is 1 way of doing it.
The tens place can now be filled by any of the remaining 5 digits (2, 3, 6, 7, 9). So, there are 5 ways of filling the tens place.
The hundreds place can now be filled by any of the remaining 4 digits. So, there are 4 ways of filling it.
Required number of numbers = (1 x 5 x 4) = 20.

406) Find the probability of making a 5-digit number from the digits 2, 3, 4, 7 and 8, in such a way that 7 is the first number and 8 is the last number?

a) 1/20	b) 20%
c) 120	d) 360

407) A 4-digit number can begin with any number except 0 and 9. What is the probability that it will start with 5 and end at 3?

a) 1/72	b) 20%
c) 1/20	d) 8/360

408) A 4-digit Pin code can begin with any number except 0 and 9. If repetition of the same digit is allowed, , the probability of Pin code beginning with 5 and ending at 3 is

a) 1/10	b) 1/100
c) 1/90	**d) 1/80**

409) A 5-digit pin code can begin with any number except 0 and 9. If repetition of the same digit is allowed, the probability of Pin code beginning with 3 and ending with a 5 is:

a) 1/10	b) 1/100
c) 1/90	**d) 1/80**

410) Following numbers are given 1, 2, 5, 6, 8, 9 you are required to make a 3-digit number which is divisible by 5. How many arrangements are possible?

Answer:

A number can only be divisible by 5 if the last digit is 5. So last digit is fix but 10^{th} place and 100^{th} place digit can be any from remaining 5 digits.

Answer: 4 x 5 x 1 = 20 ways

411) There are 5 red and 7 black cars for sale at fast wheels. If 2 cars are sold, what is the probability that both are red?

a) 0.6364	b) 0.1515	c) 0.3636	d) 0.4167

412) There are 5 red and 7 black cars for sale at fast wheels. If 2 cars are sold, what is the probability that first is red& second is black?

a) 0.6364	b) 0.1515	c) 0.2651	d) 0.7349

413) There are 6 red and 9 black cars for sale at fast wheels. If 2 cars are sold, what is the probability that first is red& second is black?

a) 0.6364	b) 0.1515

c) 0.7429	**d) 0.2571**

414) Three dices rolled together. The probability of rolling a 3 on atleast one of three dices is:

a) 0.3333	b) 0.3472
c) 0.4212	d) 0.5787

415) Three dices rolled together. The probability of rolling a 2 on atleast one of three dices is:

a) 0.3333	b) 0.3472
c) 0.4212	d) 0.5787

416) Three dices rolled together. The probability of rolling a 4,5,6 on each dice respectively is:

a) 1/216	b) 6/216
c) 3/216	d) None of these

417) Three dices rolled together. The probability of rolling a 4,5,6 on them is:

a) 1/216	**b) 1/36**
c) 3/216	d) None of these

418) Four dices are rolled together. The probability of rolling a 1,2,3,4 on them is:

a) 1/54	b) 1/27
c) 1/36	d) 1/216

419) If two dices are rolled then probability of appearing at least one six appear?

a) 0.598	b) 0.402	c) 0.694	**d) 0.306**

420) Two dice are rolled together. The probability of getting atleast one six is:

a) 2/36	b) 9/36
c) 11/36	d) 25/36

421) In a consignment of 25 auto-batteries, 3 are defective. If a random sample of 5 batteries is selected, then probability of having exactly 2 defective batteries in the sample is:

a) 5.3%	b) 3.5%
c) 12%	**d) 8.7%**

422) In a consignment of 25 Bulbs, 3 are defective. If a random sample of 5 bulbs is selected, then probability of having exactly 2 defective Bulbs in the sample is:

a) 5.3%	b) 3.5%
c) 12%	**d) 8.7%**

423) While checking out and from a departmental store, a customer passes through one out of the 12 cash counters C1 to C12 with equal probability. After that his bill is verified by one of the 3 verifying officers V1, V2 or V3 with equal probability and then he embarks on one of the two elevators E1 or E2 and is twice as likely to embark on E2 as it is twice as large as E1. What is the probability that a consumer will pass through C_1, verified by either V_1 or V_3 and embark on E_2..

a) 17/12	**b) 1/27**
c) 1/36	d) 1/54

Answer

$P(C_1) \times (V_1 \text{ or } V_3) \times (E_2)$

$$= \left(\frac{1}{12}\right)\left(\frac{1}{3}+\frac{1}{3}\right)\left(\frac{2}{3}\right)$$

424) In how many different ways can the letter of the word "BINOMIAL" be arranged in such a way that the vowels always come together?

a) 120	b) 40320
c) 2880	**d) 1440**

425) Ali wishes to plant flowers in front of his house. His father has brought him a box containing 3 tulips, 4 roses and 3 jasmines. If he selects five flowers at random, the probability that 1 tulip, 2 roses and 2 jasmines are selected is:

a) 1/9	b) 2/27
c) 3/14	d) 3/28

426) If a coin is flipped three times, the possible sample will be:

a) HHH, HTT, HTH, TTT, HTT, THH, HHT, THT
b) HTT, THT, HTH, HHH, TTH, TTT

c) HHH, THT, HTH, HTT, THH, THT, TTH, TTT	
d) HHH, TTT, THT, HTH, HHT, TTH, HTH	

427) A coin is tossed 3 times find the probability that it lands on head exactly once is?

a) 0.125	**b) 0.375**
c) 0.5	d) 0.25

428) A loaded coin is tossed 5 times and the probability of all head appear is 0.16807. what is the probability of one head tossed once?

a) 0.7	b) 0.4
c) 0.3	d) 0.6

429) A wallet contains fifty Rs. 1,000 and fifty Rs. 500 currency notes. If four notes are drawn from the wallet at random with replacement, the probability that the total amount drawn would exactly be equal to Rs. 3,000 is:

a) 25.0%	**b) 37.5%**
c) 50.0%	d) 62.5%

430) A wallet contains sixty-five Rs. 1,000 and thirty-five Rs. 500 currency notes. If four notes are drawn from the wallet at random with replacement, the probability that the total amount drawn would exactly be equal to Rs. 3,000 is:

a) 31.05%	b) 35.00%
c) 50.00%	d) 65.00%

431) In a bakery, 3 cakes of fresh cream pineapple, 4 cakes of chocolate, 2 cakes of buttercream strawberry and 1 cake without cream are available. If two customers purchase one cake each, such that first cake is replaced before the sale of the second then, the probability that both the cakes would be of chocolate flavour is:

a) 7/12	b) 2/15
c) 0.12	**d) 0.16**

432) In a bakery, 3 cakes of fresh cream pineapple, 4 cakes of chocolate, 2 cakes of buttercream strawberry and 1 cake without cream are available. If two customers purchase one cake each, such that first cake is replaced before the sale of the second then, the probability of selling both the pineapple flavour cakes is:

a) 7/12	b) 2/15
c) 0.16	**d) 0.09**

433) A problem in statistics is given to three students A, B, C whose chances of solving are ½, ¾,1/4 respectively. What is the probability that problem will be solved?

a) 3/32	**b) 29/32**
c) 3/7	d) ½

Answer

Problem will be considered as solved if anyone or more of the students will solve it.

P (Problem solve) =P (A solve or B solve or C solve) =1-P (none solve)

P(A)=1/2

	A	**B**	**C**
Solve the problem	½	¾	¼
Not solve	**1/2**	¼	¾

$$P(Solve) = 1 - P(\overline{A} \cap \overline{B} \cap \overline{C}) = 1 - \frac{1}{2} \times \frac{1}{4} \times \frac{3}{4} = 1 - \frac{3}{32} = \frac{29}{32}$$

Correct option is B

434) If two dice are rolled, what is the probability that either the sum of the two will be seven or at least one of the dice will show the number 5

a) 18/36	b) 6/36
c) 15/36	d) 12/36

435) The events A and B are independent events and P(A) = 0.5 and P(B)= 0.4, then P (A and B) is:

a) 0.9	**b) 0.20**
c) 0.54	d) 0.1

436) The events A and B are mutually exclusive. If P(A) = 0.5 and P(B) = 0.4, then P (A or B) is:

a) 0.1	b) 0.54	c) 0.9	d) 0.2

437) A and B are two mutually exclusive events and P(A) = 0.4 and P(B) 0.3 Find P(AUB)

a) 0.58	**b) 0.70**	c) 0.4	d) 0.3

438) Following data is given

Goal	0	1	2	3	4	5	>5
P(X=x)	0.05	0.2	0.15	0.15	0.3	0.05	0.1

Find the probability that there would be total 5 goals in two matches

a) 0.17	b) 0.83
c) 0.84	d) 0.12

Answer

5 goals in 2 match can be scored in following order

(i) 0 goals in first match and 5 goals in second match 0.05*0.05=0.0025
(ii) 1 goals in first match and 4 goals in second match 0.2*0.3=0.06
(iii) 2 goals in first match and 3 goals in second match 0.15*0.15=0.0225
(iv) 3 goals in first match and 2 goals in second match 0.15*0.15=0.0225
(v) 4 goals in first match and 1 goal in second match 0.3*0.2=0.06
(vi) 5 goals in first match and 0 goals in second match 0.05*0.05=0.0025

Total probability is 0.17

439) A mobile service provider offers the following options to its customers

Call metering	5 sec	20 sec	30 sec	60 sec
Fixed monthly charges	0 with no free minutes	Rs 300 with 300 free Minutes	Rs 1,000 Rs for 1,500 free Minutes	
SMS bundle charges	Rs 1 per SMS	Rs 30 for 300 SMS bundle	Rs 100 for 10,000 SMS bundle	
GPRS package	0 with no GPRS	Rs 300 Rs for 300 MB download limit	Rs 500 for 500 MB download limit	

In how many ways can a customer select a service package?

a) 10	b) 12
c) 36	**d) 108**

440) A mobile service provider offers the following options to its customers

Call metering	1 Min	2 Min	3 Min	4 Min
Call package	300 Rs for 300 Min	400 Rs for 400 Min	500 Rs for 500 Min	
GPS package	300 Rs for 300 MBs	400 Rs for 450 MBs	450 Rs for 500 MBs	

In how many ways can a customer select a service package?

a) **36**	**b)** 144	c) 16	d) none of these

Answer

No of ways = call metering option x call package options x GPS package options

No of ways = 4x3x3=36

Correct option is A

441)

Call metering	1 Min	2 Min	3 Min	4 Min
Call package	300 Rs for 300 Min	400 Rs for 400 Min	500 Rs for 500 Min	
GPS package	300 Rs for 300 MBs	400 Rs for 450 MBs	450 Rs for 500 MBs	
SMS package	500 per month	750 per month	1000 per month	

If a customer wants to choose one of these packages, then find the number of ways available to him

a) 36	**b) 108**	c) 120	d) none of these

Answer

No of ways = call metering option x call package options x GPS package options

No of ways = 4x3x3x3=108

Correct option is B

442) Eight runners have equal chance of winning the race. What is the probability that a person wins the bet if he selects top 3 runners in the correct order?

a. 5.61%	**b. 1/336**	c. 1/112	d. None of these

443) There are 12 runners in marathon and all runners have equal chance of winning. What is the probability that a person may win a bet on the race if he has to correctly select the top 3 runners and the order they finish in?

a) 1/1320	b) 1/1728
c) 3/1728	d) 1/12

Answer

Total outcomes = 12P3 = 1320

Probability = 1/1320

444) In a T20 cricket match between falcon club (FC) and eagle club (EC) the probability of winning by FC is 0.4, in a series of 5 T20 matches, the probability that FC would win exactly two matches, is:

a) 34.56%	b) 65.44%
c) 66.33%	d) 33.79%

445) A firm installed two machines U and V, on January 1, 2017. The probability that the machines will break down during first year of operations is 0.2 and 0.1 for machines U and V respectively. The probability that one of two machines will break down during the year is:

a) 0.02	**b) 0.26**
c) 0.28	d) Cannot be calculated

446) In a certain town, 70% of the households own a UPS, 30% own a generator and 20% own both a UPS and generator. The population of households that own neither a UPS nor a generator is:

a) 10%	**b) 20%**
c) 30%	d) 50%

447) In a certain town, 50% of the households own a cellular phone, 40% own a pager, and 20% own both a phone and pager. The proportion of households that own neither a cellular phone nor a pager is:

a) 90%	b) 70%
c) 10%	**d) 30%**

448) In how many different ways can the letters of the word "CORRECT" be arranged

a) 210	**b) 1260**
c) 2520	d) 5040

Answer

No of ways = 7!/2!x2! = 1260

449) A bookcase contains 6 math books and 12 accounting books. If a student randomly selects two books, then find the probability that both of them are math books or accounting books

a) 0.0423	b) 0.5556
c) 0.5000	**d) 0.5294**

450) A local news channel has conducted an opinion poll for constructing more dams in the country. The poll result indicates that 80% of the participating viewers support the idea, 10% are against the idea and 10% are undecided. If a sample of 10 participating viewers is selected at random, the probability that atleast 8 viewers will support the idea will be:

a) 0.2684	b) 0.3020
c) 0.3222	**d) 0.6778**

451) A bag contains 2 red, 3 green and 2 blue balls. Two balls are drawn at random. What is the probability that none of the balls drawn is blue?

a) 0.4762	b) 0.1429
c) 0.1905	d) 0.4000

452) A bag contains 2 red, 3 green and 2 blue balls. Two balls are drawn at random. What is the probability that one of the balls drawn is green?

a) 0.4286	b) 0.6857
c) 0.1143	**d) 0.5714**

453) There are 2 red balls, 2 green and 3 blue balls in a bag. It two balls are drawn at random. What is the probability that none is blue?

a) 0.2857	b) 0.5875
c) 0.2458	d) 0.2485

454) If a dice is rolled 4 times, then probability of appearing 3 at least one time is?

a) 0.4823	b) 0.5177	c) 0.612	d) 0.388

455) In a certain town, 50% of the households own a cellular phone, 40% own a pager, and 20% own both a phone and pager. The proportion of households that own neither a cellular phone nor a pager is_______.

456) an analysis of the frequency with which a football team scores foals in a match show that probability of securing goals in match is as follows

Goals	0	1	2	3	4	5	More than 6
probability	0.10	0.25	0.20	0.15	0.20	0.05	0.05

The probability that the team will score a total of 5 goals in a two-match series is

a) 0.17	b) 0.83	c) 0.82	d) 0.12

457) There are five red and seven black cars for sale at F wheels. If 2 cars are sold what is the probability that both are red

a) 0.6364	**b) 0.1515**
c) 0.3636	d) 0.4167

12 Probability distributions

458) If 2 coins are tossed, and the probability of appearing a head is 0.144, what is the probability that at least 1 head appears...!?

(a) 0.6	**(b) 0.3**	(c) 0.7	(d) 0.5

459) Normal distribution depends on:
1) Mean
2) Standard deviation

a) Only statement 1 is correct	b) Only statement 2 is correct
c) Both statements are correct	d) None of the above are correct

460) What is/are the parameter(s) of normal distribution?

a) Mean	b) Standard deviation	**c) Both**	d) none

461) An IQ test is administered by a well-known testing center. The test has a mean score of 100 and standard deviation of 15. If Ali's z-score is -1.20, what was his score on the test?

a) 82	b) 100
c) 112	d) 118

462) Which of the following statement as regard to normal distribution is/are correct?
i. Both tails of the distribution approach and meet the horizontal axis at a finite but high value
ii. Lower Standard Deviation leads to a flatter curve

a) Only statement 1 is correct	b) Only statement 2 is correct
c) Both statements are correct	**d) None of the above are correct**

463) A national achievement test is administered annually to 3rd graders. The test has a mean score of 100 and standard deviation of 15. If a particular students z-score is 1.20 what was his score on the test?

a) 118	b) 87
c) 92	d) 82

464) The average number of traffic accidents on a certain section of highway is two per week. Assuming that the number of accidents follow a Poisson distribution. Find the probability of 3 accidents on this section of highway during two-week period.

a) 19.5%	b) 28.9%
c) 23.8%	d) 12.8%

465) If the sample mean of a data set is 15 and the sample standard deviation is 9 what percent of the data would you expect to fall between 6 and 24 assuming that distribution is fairly symmetric

a) 81.5%	**b) 68.3%**
c) 95%	d) 99.73%

466) A sample of 4 different calculators is randomly selected from a group containing 57 calculators out of which 36 are defective. The probability that all the selected calculators are defective is:

a) 0.1400	**b) 0.1491**
c) 0.0184	d) 0

467) 8 people are selected at random from the group of 10 men and 11 women to form a committee. The probability that at least 5 men would be selected on the committee is 0.1401

a) True	**b) False**

468) If a student randomly guesses 20 multiple choice questions with four possible choices. The probability that the student would get exactly four right answers is 46.23%.

c) True	**d) False**

469) Which **TWO** of the following statements as regards the Normal Distribution are not correct?

a) Area under the Curve represents probability and so totals to 1	**b) Lower standard deviation leads to Flatter curve**
c) Both tails of the normal distribution curve always meet the horizontal axis.	d) It is described by its Mean and Standard deviation

470) The weights of bag of rice packed on a machine are normally distributed with mean 5.15kg and SD 0.05kg if a bag is picked at random probability that it weighs less than 5 kg is:

a) 0.9987	b) 0.5013
c) 0.4987	d) 0.0013

471) Which of the following statements as regards the Normal Distribution is NOT correct?

a) Both tails of the distribution approach and meet the horizontal axis at a finite but high value.
b) Higher standard deviation leads to a flatter curve
c) The area under the curve represents probability and so totals to 1.
d) It is described by its mean and standard deviation.

472) Which of the following statements as regards the Normal Distribution is correct?

a) The area under the curve represents probability and so totals to 100%
b) The Flatter the curve, the higher the standard deviation
c) The Curve is obtained by Mean(μ) and standard deviation (σ)
d) All are correct

473) An unprepared student makes random guess for ten true-false questions on a quiz. The probability of atleast one correct answer is:

a) 0.9990	b) 0.99
c) 0.9	d) 1

474) An unprepared student makes random guess for ten MCQs questions having 4 options, on a quiz. The probability of atleast one correct answer is:

a) 0.9990	b) 0.99
c) 1	d) 0.9437

475) The weights of bags of rice packed on a machine are normally distributed with mean = 5.15 kg and standard deviation 0.05kg. If a bag is picked at random, find the probability that it weighs less than 5 kg is?

a) 0.9987	b) 0.5013
c) 0.4987	d) 0.0013

476) The weights of bags of rice packed on a machine are randomly distributed with mean = 5.05 kg and standard deviation 0.02kg. if a bag is packed at random, the probability that it weighs between 5 kg and 5.06 kg is:

a) 47.72%	**b) 68.53%**
c) 34.13%	d) 49.86%

477) The weights of bags of rice packed on a machine are randomly distributed with mean = 5.05 kg and standard deviation 0.02kg. if a bag is packed at random, the probability that it weighs less than 5 kg is:

a) 0.62%	b) 2.28%
c) 2.5%	d) 49.38%

478) The weights of bags of rice packed on a machine are normally distributed with mean = 5.05 kg and standarddeviation 0.025kg. If a bag is picked at random, find the probability that it weighs less than 5.06 kg is?

a) 50.82%	b) 1.64%	c) 65.54%	d) 47.72%

479) The weights of bags of rice packed on a machine are randomly distributed with mean = 5.05 kg and standard deviation 0.025kg. if a bag is packed at random, the probability that it weighs between 5 kg and 5.06 kg is:

a) 15.54%	b) 47.72%
c) 63.26%	d) 0.2.28%

480) The weights of bags of rice packed on a machine are randomly distributed with mean = 5.05 kg and standard deviation 0.025kg. if a bag is packed at random, the probability that it weighs less than 5kgs is:

a) 15.54%	b) 47.72%
c) 63.26%	**d) 2.28%**

481) If a dice is rolled 5 times, then probability of appearing 3 at least one time is?

a) 0.598	b) 0.402
c) 0.612	d) 0.388

482) The helpline of an ISP receives an average of four calls in every five minutes during peak load hour. Assuming an appropriate Poisson distribution, what is the probability that three or more calls will be received during a period of ten minutes?

a) 0.8488	b) 0.9576
c) 0.9862	d) 0.0424

483) Fantastic Five Football (FFC) team loses 7% of its matches in every session. The probability that FFC will lose exactly one of its next five matches is?

a) 7%	**b) 26.18%**	c) 0.01%	d) 73.82%

484) Historical data shows that Star Football Club (SFC) team loses 9% of its matches in every session. The Probability that SFC will lose exactly one of its next five matches is:

a) 04.50%	b) 09.00%
c) 08.50%	**d) 30.86%**

485) PABX system of Nadia Travels receives an average of two calls in every three minutes. Assuming Poisson distribution, what is the probability that four or more calls will be received during a period of 9 minutes?

a) 0.1512	b) 0.2851
c) 0.6667	**d) 0.8488**

486) PABX system of Nadia Travels receives an average of two calls in every three minutes. Assuming Poisson distribution, what is the probability that six or more calls will be received during a period of 9 minutes?

a) 0.4457	**b) 0.5543**
c) 0.6063	d) 0.8488

487) You are given 7 true false questions and required to find the probability of at least 1 will be correct?

a) 0.992	**b)** 0.008
c) 0.9375	d) 0.0625

488) In a test there are 20 questions, all questions are MCQs with 4 options one of which is correct. 4 questions are selected at random from the questions, find the probability that all are correct:

a) 18.97%	b) 20.23%
c) 8.67%	d) 33.75%

489) In a test there are 10 questions, all questions are MCQs with 4 options one of which is correct. The student only knows the answer of three questions and 5 questions are needed to pass the test, find the probability that he will pass:

a) 75.64%	b) 44.49%
c) 8.67%	**d) 55.51%**

490) In a test there are 20 questions, all questions are MCQs with 4 options one of which is correct. He solved all question haphazardly, find the probability that exactly 4 are correct:

a) 18.96%	b) 19.86%
c) 16.98%	d) 16.66%

491) A box contains 2 white balls, 3 black balls and 4 red balls. In how many ways balls be drawn from the box, if at least one black ball is to be included in the draw:

a) 32	b) 48
c) 64	d) 96

492) A telephone operator receives on average 2 calls in 3 minutes. Find the probability of receiving more than or equal to 4 calls in 9 minutes:

a) 0.5120	**b) 0.8488**
c) 0.5217	d) 0.4884

493) Ahmed received on average 4 calls in 5 minutes. Find the probability of receiving more than 3 calls in 10 minutes:

a) 0.0286	b) 0.2682
c) 0.2862	d) 0.0862

494) During peak hours a center receives 4 calls per 30 minutes. What is the probability of getting 3 or more calls in an hour?

a) 0.9862	b) 0.9682
c) 0.8692	d) 0.2689

495) A consignment of 12 refurbished CNG kits contain 4 defective kits. If 4 Kits are selected at random then find the probability that at least 3 are defective

$P(x \geq 3) = P(x=3) + p(x=4)$

$$\frac{{}^{4}C_3 \times {}^{8}C_1 + {}^{4}C_3 \times {}^{8}C_1}{{}^{12}C_4} = \frac{1}{15}$$

496) A consignment of 12 refurbished CNG kits contain 4 defective kits. If 5 Kits are randomly chosen for inspection, what is the probability that at least 3 of them are defective?

$P(x \geq 3) = P(x=3) + p(x=4)$

$$\frac{{}^4C_3 \times {}^8C_2 + {}^4C_4 \times {}^8C_1}{{}^{12}C_5} = \frac{5}{33}$$

a) 1/11	b) 2/33
c) 5/33	d) None of these

497) A committee is to be made of 4 members from a total of 13 people of which 5 from Punjab, 2 fromBaluchistan, 4 from Sindh and 2 from KPK, what is the probability that committee contain none from Punjab?

a) 6.8%	b) 0.7%	**c)** 9.8%	d) None of these

498) A committee is to be made of 5 members from a total of 12 people of which 5 from Punjab, 2 from Baluchistan, 3 from Sindh and 2 from KPK, what is the probability that committee contain 2 from Sindh and 1 each from others?

a) 6.8%	b) 0.7%	**c)** 9.8%	d) None of these

499) Sadiq earned a total of 940 on a general knowledge test. The mean test score was 850 with a standard deviation of 100. What proportion of students had a higher score than Sadiq? (assume that test scores are normally distributed)

a) 82%	b) **18%**
c) 10%	d) 32%

500) Online booking system of food channel receives an average of two orders in every four minutes. Assuming an approx. Poisson distribution that five or more orders will be received during a period of eight minutes.

a) 0.3712	b) 0.4679
c) 0.5321	d) 0.4679

501) FX food channel receives an average of six orders in every four minutes. Assuming an approximate Poisson distribution that five or more orders will be received during a period of four minutes

a) 0.2849	**b) 0.7149**	c) 0.44550	d) 0.5545